



THYNK

The Hub for Young Next-gen Knowledge

ANCIENT WISDOM. MODERN TECHNOLOGY.
— Limitless Possibilities. —

FOCUS
CREATIVITY
CONFIDENCE
CRITICAL THINKING

$\sqrt{x}$ ∞ 123
+ × ÷ { } </>

VEDIC MATH
LOGIC & REASONING
CODING
PROBLEM SOLVING
MENTAL DEVELOPMENT

Empowering Young Minds with Vedic Wisdom
and Technology for a Smarter Tomorrow.

VEDIC MATH
CODING
LOGIC & REASONING
MENTAL DEVELOPMENT
FUTURE READY

The illustration features a young boy and girl sitting at a desk. The boy is pointing upwards while looking at a laptop that has the THYNK logo on its lid. The girl is writing in a notebook. On the desk, there is a wooden abacus and a stack of five books with titles: 'VEDIC MATH', 'LOGIC & REASONING', 'CODING', 'PROBLEM SOLVING', and 'MENTAL DEVELOPMENT'. A small potted plant sits next to the books. In the background, a large blue silhouette of a head is filled with various mathematical and technological symbols like $\sqrt{x}$, ∞ , 123, +, ×, ÷, { }, </>, and a lightbulb. To the left of the head, four icons represent 'FOCUS' (target), 'CREATIVITY' (lightbulb), 'CONFIDENCE' (person with star), and 'CRITICAL THINKING' (brain). The entire scene is set against a light blue background with decorative elements like a sun, a paper airplane, and a gear.



Preface

THYNK · Level 1 · Young Explorer

The Hub for Young Next-gen Knowledge

Every generation shapes the future, but today's children are growing up in a world where technology is part of life from the very beginning. They are naturally curious, creative, and eager to explore. At **THYNK**, we believe these young minds deserve an education that nurtures not only knowledge but also the ability to think, innovate, and solve problems with confidence.

THYNK is a vision to prepare the next generation for a rapidly evolving world. Our carefully designed programmes focus on strengthening mental processes, encouraging logical thinking, and developing the skills needed to thrive in the age of technology and artificial intelligence.

Our age-appropriate courses are thoughtfully tailored to complement classroom learning while going beyond traditional teaching methods. By blending strong academic foundations with creativity, innovation, and practical application, we help every learner unlock their true potential.

At the heart of THYNK lies a simple philosophy: **learning should be enjoyable, meaningful, and progressive**. Every new concept is introduced only after reinforcing previously learned ideas, helping children build confidence step by step. This structured approach satisfies their natural curiosity, encourages independent thinking, and makes learning both engaging and rewarding.

This workbook has been written in a simple, clear, and child-friendly language, making every concept easy to understand and enjoyable to learn. Through carefully designed activities, examples, and practice exercises, we hope to inspire every learner to think deeper, explore further, and discover the joy of learning.

Welcome to **THYNK** - where young minds learn to **Think, Create, and Innovate**.

YOUR MAP FOR THE YEAR



What's **Inside** My Workbook?

Seven exciting topics that turn you into a real coder — one block, one sutra, one project at a time.

1

Introduction

Meet your new playground — sprites, stage, blocks & the magic coding screen.

2

Sequencing & Events

Make things happen in the right order — click, press a key, watch the magic begin.

3

Loops

Repeat and *forever* — paired with the Vedic sutra **Ekadhikena Purvena**.

4

Variables & Lists

Tiny boxes that remember numbers, names and scores for your games.

5

Conditionals (if / else)

Teach the computer to make smart choices — just like you do every day.

6

Operators & Maths Blocks

Add, subtract, multiply — and bring Vedic Maths to life inside your code.

★

Mini-Projects at the End of Every Chapter

Each big topic ends with a fun project — a dancing cat, a counting caterpillar, a number-guessing game and many more — so you don't just *learn* coding, you *build* things.

✍️

Exercises & Self-Practice

One full page of fun exercises for every topic — predict, fix the bug, build it yourself. The best way to **really** learn coding is to *do* it on your own!

★

Bonus Capstone · Water Cycle Animation

The grand finale! Use *everything* you've learned — events, broadcasts, variables, loops **and** cloning — to build a real **Water Cycle** science animation. Take it home, show your family, and watch evaporation, condensation and rain happen on your screen.

How to use this workbook: Read the colourful boxes, try the blocks on screen, finish the activities, and tick the "I did it!" box at the end of every topic. Keep this book safe — by the end of the year it will be full of *your* ideas and *your* code.



For the School & Teacher

GRADE 4
BEGINNERS BASIC 1 · SESSION PLAN



Suggested Session Plan · Grade 4 · Beginners Basic 1

This volume is designed as a **relaxed, project-first** introduction to Scratch coding and Vedic Maths for 9–10 year-olds. We recommend spreading it across approximately **24 class sessions of 35 minutes each**.

Chapter-wise class breakdown

#	Chapter / Module	No. of Classes	What gets covered
1	Introduction to Vedic/Scratch Vedic Maths, Meet the IDE, blocks & the A.I. Buddy	2	Vedic Maths, Scratch screen tour, block families.
2	Sequencing & Events + Vedic Warm-up: Number Patterns	4	Concept + 4 pattern practice sheets + Greeting Cat + Exercises.
3	Loops + Vedic Sutras: Ekadhikena & Purvena	4	Repeat / forever + 2 sutra worksheets + Draw-a-Square mini-project.
4	Variables & Lists Boxes that remember + Vedic Quiz skeleton	3	Set / change, lists, Coin Counter + Vedic Quiz Part 1 mini-project.
5	Conditionals (if / else) Comparisons & Quiz Checker	3	Yes/No thinking, green operators, Quiz Checker upgrade.
6	Operators & Maths Blocks Vedic Add, Subtract, $\times 11$, $\times 99$, $\div 5/25/50$	4	6 Vedic practice sheets (~180 sums) + Nikhilam & Urdhva in Scratch.
-	Exam 1 · Mid-term Assessment Chapters 1–3 · written + on-screen	2	Concept MCQs + Predict / Fix-the-Bug + short Scratch task.
-	Exam 2 · Final Assessment Chapters 4–6 + Capstone submission	2	Vedic sums + variables/if-else + short Scratch task.
Total Classes for Volume 1		24	≈ full one term (weekly)

A note for the class teacher



How to pace the year

Each chapter is designed as a **self-contained mini-unit**.
Concept pages → Vedic warm-up → mini-project → exercises → answer key. Plan **~20 minutes concept +**



Mini-projects are Homework

All three mini-projects (*Draw a Square*, *Vedic Quiz*, *Quiz Checker*) are marked with a red **Homework** badge in the workbook. Children build these **at home** and demo

hands-on and **~15 minutes exercises** per session.

them in the next class — this doubles engagement and parent involvement.

Bonus Capstone

The **Water Cycle Animation** at the end is a take-home **capstone project** tying together every concept. Plan a **show-and-tell session** or a small school-level exhibition to celebrate.

Vedic + Coding

Every chapter deliberately pairs a Scratch concept with a Vedic sutra — so students strengthen mental-maths and computational-thinking *at the same time*. Don't skip the warm-ups!

FLEXIBILITY

Sessions are indicative. A class comfortable with computers may finish a chapter faster; a class new to typing may need an extra period on Chapter 1. Feel free to reshuffle — but keep the **concept** → **warm-up** → **mini-project** → **exercise** flow inside each chapter for best learning outcomes.

CHAPTER ONE

01

Introduction

*Your colourful playground where blocks click together like LEGO —
and bring sprites to life.*

VEDIC MATHS

THE STAGE

SPRITES

BLOCK PALETTE

AI

EXERCISE



1.1 Introduction to Vedic Maths

CHAPTER 1 INTRODUCTION



Big Idea

Vedic Maths is a fast and smart way of thinking with numbers. It comes from ancient Indian knowledge and uses short rules (called **sutras**) to solve sums quickly and clearly.

What is Vedic Maths?

Vedic Maths is not about memorizing long steps. It is about **spotting patterns, breaking big problems into small pieces**, and using simple mental tricks. That is exactly how good coders think too.

Pattern Thinking

Find the rule in numbers. If the rule repeats, your brain works faster.

One Step at a Time

Do small easy steps in the right order instead of one giant difficult step.

Speed with Clarity

You get quick answers, but still understand *why* the answer is correct.

Why we are learning it with Scratch

IN VEDIC MATHS

Use a rule and repeat it

For many sums, the same tiny idea is used again and again.



IN SCRATCH CODING

Use blocks, loops, and logic

Code also repeats useful rules using loops, variables, and conditionals.

WHAT IS A SUTRA?

A **sutra** is a short math rule written in words. In this workbook, every sutra will be taught with examples first, then used inside Scratch projects.

Sutras you will meet in this book

1

Ekadhikena Purvena

"By one more than the previous" - useful for counting patterns and loops.

2

Nikhilam

"All from 9 and the last from 10" - helpful for quick subtraction and multiplication tricks.

3

Urdhva-Tiryagbhyam

"Vertically and crosswise" - a smart method for multiplication.

TRY IT! · 5 MINUTES**Pattern Warm-up**

1. Complete this pattern: **5, 10, 15, 20,** ____
2. Now write the rule in words: _____
3. Say how this is like coding in Scratch: _____

REMEMBER

Great coders are great pattern-finders. Vedic Maths trains your brain to see patterns quickly - and that makes coding easier.



1.2 The Scratch

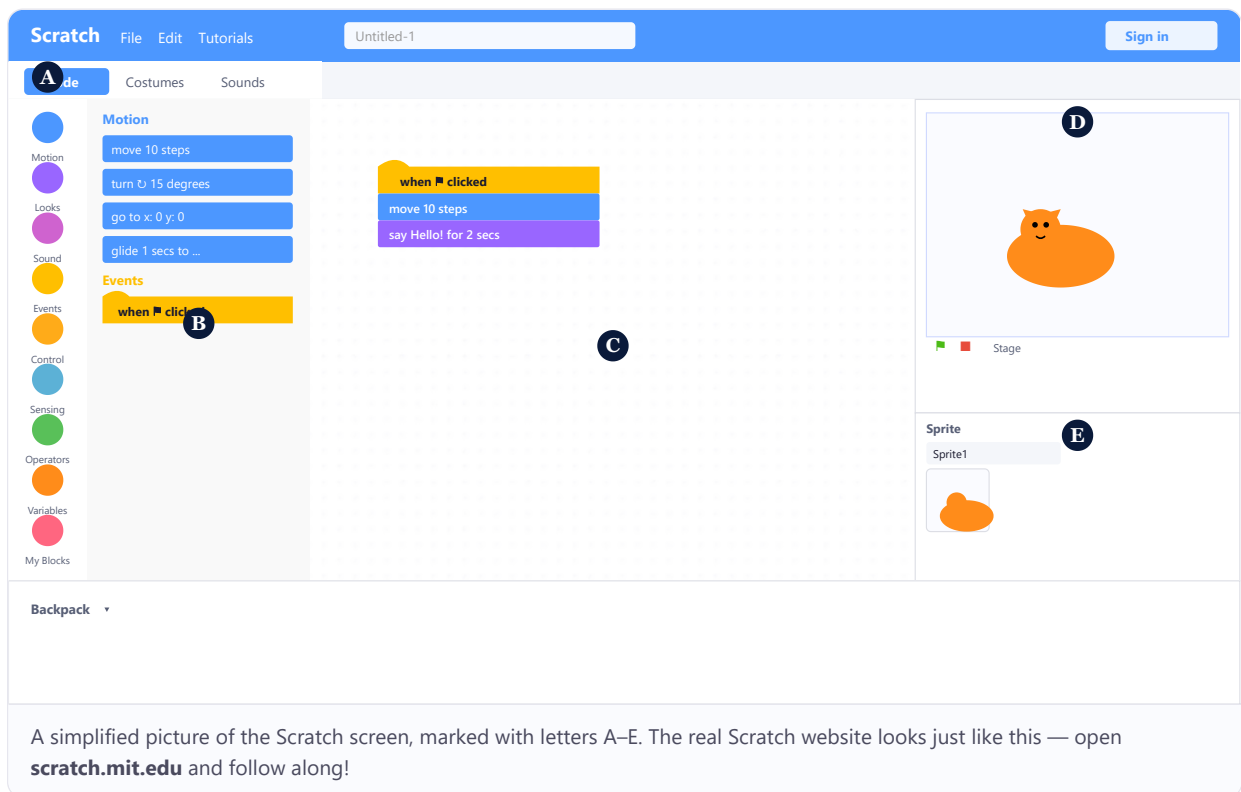
CHAPTER 1 INTRODUCTION



Big Idea

Scratch is a **free coding tool** made for children, by Massachusetts Institute of Technology (MIT is a private research university in Cambridge, Massachusetts, United States.). Open **scratch.mit.edu** in any browser and you're ready to code — no installation, no typing tricky words.

Below is a friendly map of the Scratch screen. The five letters **A** to **E** show the five most important areas — learn these and you can find everything!



A Block Categories — coloured circles like *Motion*, *Looks*, *Sound*, *Events*... Click one to see its blocks.

C Code Area — your workspace. Drag blocks here and snap them together like LEGO.

E Sprite List — every character in your project lives here. Add new ones with the cat-icon button.

B Block Palette — the box of colourful blocks you can drag out and use.

D Stage — the white screen where your sprites move. Click the green flag to start.



The 9 Colours of Scratch Blocks

CHAPTER 1 INTRODUCTION

Every block in Scratch belongs to a coloured family. Once you remember the colours, you'll know where every block lives!

Motion Move, turn, glide. Anything to do with movement . move 10 steps	Looks Say things, change costumes, grow or shrink. say Hi! for 2 secs	Sound Play music & sound effects, change the volume. play sound Meow
Events "When something happens..." — the start of every script. when clicked	Control Loops & choices — <i>repeat, forever, if...then</i> . repeat 10	Sensing Touch, ask, hear — your sprite's senses. touching edge ?
Operators Maths & comparisons — add, subtract, < >. 5 + 3	Variables Boxes that remember numbers, scores & names. change score by 1	My Blocks Make your <i>own</i> blocks — your private super-powers! jump high

Class room activity
 Color activity taken by teacher.

Three tabs above the block palette

Code The default tab. This is where you build your scripts.	Costumes Change how your sprite <i>looks</i> — draw, colour, even add a moustache.	Sounds Add or record sounds — meows, claps, your own voice.
---	--	---

TRY IT! · 5 MINUTES

Hello, Scratch!

1. Open scratch.mit.edu and click **Create**.
2. Find the yellow **Events** category. Drag **when clicked** into the Code Area.
3. From purple **Looks**, drag **say Hello! for 2 seconds** and snap it under the yellow block.
4. Click the green flag at the top of the Stage. Did the cat speak? 🐱



1.3 Meet Your A.I. Buddy

BONUS
SMART HELPER



Stuck on a block? Just ask!

From here on, every chapter has a purple "Ask A.I." box. It shows you exactly what to type into a chat A.I. (like ChatGPT, Gemini or Copilot) when you get stuck — so you never feel lost.

What is A.I., really?

A.I. (Artificial Intelligence) is like a super-smart computer friend that has learned from millions of books, lessons, and Scratch projects. It can explain things in simple words, suggest the right coding blocks, and help you spot small mistakes. The better you ask your questions, the better it can help—just like a patient teacher who is always ready to learn with you.

3 Golden Rules of Asking A.I.

1

Be Clear

Say **what you want** the sprite to do — not just "fix it". Example: *"I want my cat to spin twice when I press space."*

2

Be Specific

Tell A.I. **which blocks you tried** and **what went wrong**. The more details, the better the help.

3

One Step at a Time

Ask about **one problem per chat**. Don't pile three questions together — A.I. helps best when focused.

COMPARE THESE TWO PROMPTS

my code doesnt work fix it

In Scratch, I want my cat sprite to say "Hi!" for 2 seconds and then move 100 steps when I press the green flag. I tried using "say Hi" and "move 100" but the cat moves first and then says Hi. How do I fix the order?

See the difference? The second prompt tells A.I. **what** you want, **what you tried**, and **what's wrong** — so it can give a real answer.

The Golden Rule — Always **try yourself first**. Drag the blocks, watch what happens, think for a minute. Only ask A.I. when you've genuinely tried and got stuck. A.I. is your *helper*, not your homework-doer. The more you struggle a little, the smarter **you** get. 🗨️ ⭐



1.4 Exercise

Number Patterns · Practice Sheet 1

CLASSWORK

CHAPTER 2

VEDIC WARM-

UP



10 quick pattern puzzles — find the rule, pick the answer!

Look at two neighbours, spot the jump, check it on another pair — then circle the right option. Total questions: **10**. Suggested time: **15–20 mins**.

1 Next in 9, 18, 27, 36, 45, ____

- a. 54 b. 76
c. 63 d. 81

2 6, 9, 12, 15, 18, 21 — what is the rule?

- a. $\times 6$ b. $+ 3$
c. $\times 4$ d. $\times 5$

3 Missing blanks in 69, 60, 51, ____, 33, ____

- a. 42, 24 b. 45, 21
c. 40, 22 d. I don't know

4 Rule of 65, 62, 59, 56, 53, 50

- a. Subtract 2 b. Subtract 3
c. Subtract 4 d. Add 3

5 Next in 11, 9, 7, 5, ____

- a. 7 b. 4
c. 3 d. 2

6 Magic Box — what is the rule?

Column A	Column B
2	6
3	9
4	12
5	?
6	?
7	?

- a. Add 4 b. Add 6
c. Multiply by 2 d. Multiply by 3

7 Next in 55, 60, 65, ____, 75, 80

- a. 70 b. 66
c. 40 d. 75

8 Rule of 99, 97, 95, 93, 91

- a. Subtract 1 b. Add 2
c. Subtract 2 d. Add 3

9 Missing in 23, 33, ____, 53, ____, 73

- a. 32, 62 b. 42, 63
c. 43, 63 d. 42, 62

10 Rule of 32, 36, 40, 44

- a. add 4 b. multiply 2
c. subtract 4 d. divide by 2

Tip — If neighbours **grow**, the rule is *add* or *multiply*. If they **shrink**, it's *subtract* or *divide*. Always check your rule on **two pairs** before you commit!

Number Patterns · Practice Sheet 2

HOMEWORK
CHAPTER 2
VEDIC WARM-
UP


Pattern Puzzles — Numbers and shapes!

A mix of number sequences, dot & block patterns, and repeating shapes.

- 1 Rule of 5, 10, 15, 20, 25**
- a. add 6 b. add 5
c. add 4 d. add 7

- 2 Missing in 18, 24, __, 36, 42**
- a. 28 b. 31
c. 30 d. 35

- 3 Missing in 22, 29, 36, __, 50, __**
- a. 35, 55 b. 40, 57
c. 41, 56 d. 43, 57

- 4 How many counters come next?**
- 
- a. 5 b. 7
c. 9 d. 11

- 5 Next letter: A B C A B C A B C __**
- a. A b. B
c. C

- 6 How many blocks in the 4th term?**
- 
- a. 16 blocks b. 15 blocks
c. 14 blocks d. End of pattern

- 7 Missing in 69, 60, 51, __, 33, __**
- a. 42, 24 b. 45, 21
c. 40, 22 d. I don't know

- 8 Missing in 120, 115, 110, __, 100, 95**
- a. 105 b. 115
c. 90 d. I don't know

- 9 Missing in 230, 219, 208, 197, __, 175**
- a. 187 b. 186
c. 195 d. I don't know

- 10 Start with 1, multiply the previous by 3. 1, __, __, __, __**
- a. 1, 3, 6, 9, 12 b. 1, 4, 7, 10, 13
c. 1, 3, 9, 12, 15 d. 1, 3, 9, 27, 81

- 11 11th figure in this pattern?**
- 
- a.  Diamond b.  Hexagon c.  Trapezoid

- 12 Which pattern follows this rule? Start with 2, add 4 each time until there are 5 numbers.**
- a. 2, 4, 8, 12, 16 b. 2, 6, 10, 14, 17
c. 2, 6, 10, 14, 18 d. 2, 6, 11, 15, 19

Tip — For a **shape** or **letter** pattern like A B C A B C..., count how many items repeat (the **cycle**) and then divide the position by that cycle — the remainder tells you which shape/letter you'll see!

CHAPTER TWO

02

Sequencing & Events

First, second, third... coding is all about doing things in the right order — and starting them at the right moment.

SEQUENCE

WHEN ■ CLICKED

KEY PRESS

FIRST SCRIPT!

EXERCISE



2.1 What is a Sequence?

CHAPTER 2 SEQUENCING & EVENTS



Big Idea

A **sequence** is a list of steps done **one after the other**, in a fixed order. Every program ever written is just a smart sequence of small steps.

Your morning is a sequence

Think about how you get ready for school. Without even thinking, you follow a sequence:

1

Wake up

Alarm rings → you open your eyes.

2

Brush your teeth

Open tap → squeeze toothpaste → brush → rinse.

3

Wear your uniform

Shirt first, **then** tie. Not the other way around!

4

Eat breakfast & leave

Eat → wear shoes → pick bag → walk to bus.

WHAT HAPPENS IF ORDER CHANGES?

Imagine wearing your **shoes before** your socks, or eating breakfast *after* brushing your teeth. Funny — but it doesn't work, right? **Order matters**. The same is true in code.

In Scratch, blocks *snap* in sequence

When you stack Scratch blocks **from top to bottom**, they run in that exact order. The top block runs first, then the next, then the next.



Block stack

CODE AREA · TOP → BOTTOM

when clicked

move **50** steps

say **Hi friend!** for **2** secs

play sound **Meow** until done

🗣️ In simple words

1. Wait till someone clicks the green flag.
2. **First**, walk 50 steps forward.
3. **Then** say "Hi friend!" for 2 seconds.
4. **Finally**, play the meow sound.

REMEMBER

Swap any two blocks and the result **changes**. Coders must be careful planners!



2.2 Vedic Warm-up · Number Sequences

CHAPTER 2
VEDIC WARM-UP

Big Idea · Find the Secret Rule



A **number sequence** is a row of numbers that follows a **secret rule** — like "*add 2 each time*" or "*minus 5 each time*". Spot the rule and you can predict any missing number — that is exactly what a computer loop does!

HOW TO SPOT THE RULE — IN 3 EASY STEPS

1. Take any two neighbours — ask "*what did we add or subtract?*"
2. Check the same jump on a second pair.
3. If the jump is the **same every time**, that is your rule — apply it to fill any blank.

Example — **3, 6, 9, 12, ___** : $6-3=3$, $9-6=3 \Rightarrow$ rule = **add 3**, next = **15**. Try minus too: **20, 17, 14, ___, 8** $\Rightarrow$ rule = **minus 3**, missing = **11**.

Your turn · Circle the correct answer

1 Rule of 2, 4, 6, 8, 10, 12, 14

- a. Add 1 b. Add 2

2 Next in 4, 8, 12, ___

- a. 10 b. 16
c. 18

3 Rule of 18, 15, 12, 9, 6, 3, 0

- a. Add 3 b. Plus 4
c. Minus 3 d. Minus 4

4 Missing in 12, ___, 32, 42, 52

- a. 22 b. 23
c. 24

5 Rule of 103, 107, 111, 115, 119, 123

- a. Add 100 b. Add 104
c. Add 4 d. Minus 4

6 Noora starts at 13, subtracts 4 each time. Next 3 numbers?

- a. 14, 17, 20 b. 9, 5, 1
c. 12, 11, 10 d. 9, 4, 0

7 Next in 150, 155, 160, 165, ___

- a. 170 b. 175
c. 180

8 Rule of 100, 200, 300, 400, 500

- a. Add 100 b. Add 10
c. Minus 10

9 Missing in 58, 51, 44, ___, 30

- a. 42 b. 37
c. 35

10 Next in 180, 140, 100, 60, ___

- a. 40 b. 50
c. 30 d. 20

Figure out the pattern · Write the missing number

11 12, 15, ___

Answer: _____

12 4, 8, 12, 16, 20, ___

Answer: _____

13 ___, 10, 15, 20, 25

Answer: _____

14 2, 4, 6, 8, 10, ___

Answer: _____

BRIDGE TO SCRATCH

Every rule you spotted — "*add 2*", "*minus 3*" — is the recipe for a Scratch loop with a counter: **change counter by 2**, **change counter by -3**. Master the pattern here (and a couple more practice sheets coming up!) and the **Events** section will feel natural.



2.2A Vedic Worksheet · Tables 1 to 99

CHAPTER 2
VEDIC WARM-UP



Multiplication Practice Sheet

Practice Vedic multiplication tables from **1 to 9**. Read each table aloud, then solve the worksheet carefully.

1.

$$\begin{array}{r} 85 \\ \times \quad 3 \\ \hline \\ \hline \end{array}$$

2.

$$\begin{array}{r} 87 \\ \times \quad 4 \\ \hline \\ \hline \end{array}$$

3.

$$\begin{array}{r} 88 \\ \times \quad 7 \\ \hline \\ \hline \end{array}$$

4.

$$\begin{array}{r} 99 \\ \times \quad 9 \\ \hline \\ \hline \end{array}$$

5.

$$\begin{array}{r} 23 \\ \times \quad 4 \\ \hline \\ \hline \end{array}$$

6.

$$\begin{array}{r} 54 \\ \times \quad 8 \\ \hline \\ \hline \end{array}$$

7.

$$\begin{array}{r} 83 \\ \times \quad 6 \\ \hline \\ \hline \end{array}$$

8.

$$\begin{array}{r} 56 \\ \times \quad 9 \\ \hline \\ \hline \end{array}$$

9.

$$\begin{array}{r} 77 \\ \times \quad 5 \\ \hline \\ \hline \end{array}$$

10.

$$\begin{array}{r} 45 \\ \times \quad 2 \\ \hline \\ \hline \end{array}$$

11.

$$\begin{array}{r} 34 \\ \times \quad 1 \\ \hline \\ \hline \end{array}$$

2.

$$\begin{array}{r} 11 \\ \times \quad 9 \\ \hline \\ \hline \end{array}$$

13.

$$\begin{array}{r} 75 \\ \times \quad 8 \\ \hline \\ \hline \end{array}$$

14.

$$\begin{array}{r} 98 \\ \times \quad 7 \\ \hline \\ \hline \end{array}$$

15.

$$\begin{array}{r} 34 \\ \times \quad 6 \\ \hline \\ \hline \end{array}$$

16.

$$\begin{array}{r} 76 \\ \times \quad 5 \\ \hline \\ \hline \end{array}$$

17.

$$\begin{array}{r} 45 \\ \times \quad 2 \\ \hline \\ \hline \end{array}$$

18.

$$\begin{array}{r} 87 \\ \times \quad 3 \\ \hline \\ \hline \end{array}$$

19.

$$\begin{array}{r} 67 \\ \times \quad 9 \\ \hline \\ \hline \end{array}$$

20.

$$\begin{array}{r} 13 \\ \times \quad 7 \\ \hline \\ \hline \end{array}$$

21.

$$\begin{array}{r} 75 \\ \times \quad 6 \\ \hline \\ \hline \end{array}$$

22.

$$\begin{array}{r} 56 \\ \times \quad 1 \\ \hline \\ \hline \end{array}$$

23.

$$\begin{array}{r} 34 \\ \times \quad 5 \\ \hline \\ \hline \end{array}$$

24.

$$\begin{array}{r} 34 \\ \times \quad 8 \\ \hline \\ \hline \end{array}$$

CLASS + HOME PRACTICE

How to use this worksheet

1. Revise each multiplication table from 1 to 9.
2. Solve the worksheet once in class and once at home.
3. Try to improve speed and accuracy in the second attempt.



2.3 What is an Event?

CHAPTER 2 SEQUENCING & EVENTS

DEFINITION An **event** is something that **happens** — like a click, a key being pressed, or a message being sent. Events are the **starting whistle** for our sequences.

Yellow "Hat" blocks live in the Events family

Event blocks are shaped like a *hat* — round on top. They **sit at the top** of every script. Nothing snaps above them, because nothing comes *before* the start!

Green Flag

when clicked

Runs when you click the green flag above the stage.

Key Press

when space key pressed

Runs the moment you press the chosen key.

Sprite Click

when this sprite clicked

Runs when you click *this* sprite on the stage.

Sample Project · "Greeting Cat"

Goal: When you click the green flag, the cat walks and greets you. When you press **space**, it dances by spinning.

Script 1 · Greeting

WHEN THE GREEN FLAG IS CLICKED

when clicked

go to x: **-150** y: **0**

glide **2** secs to x: **0** y: **0**

say **Hello, I'm Scratchy!** for **2** secs

Script 2 · Dance

WHEN THE SPACE KEY IS PRESSED

when space key pressed

turn **360** degrees

play sound **Meow** until done

What you'll see on the Stage The Hub for Young Next-gen Knowledge

Stage



After clicking — the cat glides to the centre and greets you. Press **space** and watch it spin!

YOUR TURN · ACTIVITY 3.A

Make Scratchy reply with your name

1. Open the Greeting Cat project above.
2. Change the speech text from "Scratchy" to your own name.
3. Add a third script that runs **when this sprite clicked** and makes the cat say "Ouch!" for 1 second.
4. ☐ I did it!



ASK A.I. Stuck? Try this prompt —

I'm coding in Scratch. I want my cat sprite to **say "Hi, friend!" for 2 seconds and then move 80 steps** when I click the green flag. I tried two blocks but the cat moves *before* it speaks. Which **Events** block should I start with, and what's the correct **order** of blocks so it talks first, then walks?

💡 Notice how the prompt names the **topic** (sequencing & events) and tells A.I. **what went wrong**.



2.4 Exercises · Sequencing & Events

CHAPTER 2 PRACTICE



Get the order right!

Three challenges to test your sequencing super-power. Pencil ready — let's go.

1 PREDICT What does the cat do?

Read this stack from top to bottom. In which order will things happen on the stage? Write 1, 2, 3, 4 in the boxes on the right.

```

when clicked
  move 30 steps
  say Hello! for 1 sec
  play sound Meow until done
  turn 15 degrees
  
```

Order	What happens
	Cat turns 15°
	Cat says "Hello!"
	Cat moves 30 steps
	Cat plays meow sound

2 FIX THE BUG Wrong order!

Riya wants her cat to **say "Hi" first, then jump** (move 50 steps). But she wrote the blocks the wrong way round. **Cross out** the wrong stack and **rewrite the correct one** in the empty box.

```

when clicked
  move 50 steps
  say Hi! for 1 sec
  
```

3 BUILD IT! The Birthday Sprite 🎂

Build this in Scratch:

- When the **space key** is pressed, the cat should say **"Happy Birthday!"** for 2 seconds.
- Then it plays a **sound** of your choice.
- Then it says **your friend's name** for 2 seconds.

Which **Events** block will start your script? Write its name:

☐ I built and tested my Birthday Sprite!

Tip — The block at the **top** always runs first. The order of the rest *changes the story*.

Number Patterns • Practice Sheet 1

HOMEWORK

CHAPTER 2

VEDIC WARM-
UP

16 quick puzzles — hop, skip & back-count!

Count in 10s, 20s, 50s, 100s and 1,000s — and use **1 more** / **1 less**. Total: **16 Qs** · Suggested time: **16 mins**.

- 1** Next in the shape pattern:



- a.  Purple right-tri
- b.  Blue triangle
- c.  Yellow triangle
- d.  Red right-tri

- 2** Missing in 20, 30, ____, 50, 60

- [illegible]

- 3** Next in 320, 340, 360, _____

- a.** 54 **b.** 400
c. 380 **d.** 370

- 4 Missing in 150, 200, ____, 300

- a. 250 b. 225
c. 275 d. 350

- 5** Count back by 10s: 90, ____, ____, ____

- a. 80, 70, 60 b. 80, 70, 50
c. 100, 110, 120

- 6** Missing in 520, , 560, 580,

- a.** 540, 590 **b.** 540, 600
c. 525, 590 **d.** 510, 590

- 7 Missing in 600, 550, ____, ____, 400, 350, 300, 250, 200

- a.** 500, 400 **b.** 350, 500
c. 450, 500 **d.** 500, 450

- 8** Missing in 1,403, ____, 3,403, 4,403

- a.** 1,403 **b.** 2,000
c. 2,503 **d.** 2,403

- 9** Next number: 1,680, 1,780, 1,880,

Answer: _____

- 10** Next number: 6,357, 6,358, 6,359,

Answer: _____

- 11** 5,447 is 1,000 more than 4,447.

- a.** True **b.** False

- 12** 1 more than 8,902 is ____

- a.** 8,901 **b.** 8,903
- c.** 8,900

- 13** 1 less than 3,450 is ____

- a.** 3,449 **b.** 3,451
- c.** 3,452

- 14** 10 less than 2,475 is 2,485.

- a.** True **b.** False

- 15** ___ is 10 less than 650.

- a. 670 b. 660
- c. 640

- 16** 6,046 is ___ less than 7,046.

- a. 1,000 b. 100
- c. 10

Tip — For “1 more” / “1 less”, only the last digit changes. For “10 more” / “10 less”, only the tens digit changes. For “100 more”, only hundreds. Line up the place values in your head!

Number Patterns · Practice Sheet 2

HOMEWORK



10 Input & Output puzzles — find the rule!

A **rule** tells the machine how to turn every **input** into an **output**. Spot it once, and every row falls into place. Total: 10 Qs · Suggested time: 13 mins.

1 What real-life thing do we use as a picture for the input & output table?

- a. A soda machine b. A television
c. Shoes d. A transformer

2 What never changes?

- a. The rule b. The output
c. The input d. The pattern

3 What is the output?

◇	♥
10	15
15	20
20	25
25	?

- a. 35 b. 40
c. 30 d. 50

4 What is the output?

◇	♥
900	800
700	600
500	?
300	200

- a. 300 b. 600
c. 100 d. 400

5 Correct values for the outputs? (pick two)

◇	♥
10	21
12	23
13	?
14	?

- a. 22 b. 24
c. 10 d. 25

6 Correct values for the outputs? (pick two)

◇	♥
550	50
450	400
350	?
150	?

- a. 300 b. 250
c. 200 d. 100

7 Rule (number expression)?

★	Rule	☺
17	?	24
18	?	25
27	?	34
45	?	52

- a. +8 b. $\times 6$
c. +7 d. $\times 5$

8 Fill in the missing input & output. (pick two)

★	☺
1	9
3	11
5	13
?	15
9	?

- a. Input 6 b. Input 7
c. Output 17 d. Output 16

9 Number Machine rule (input → output)?

Input	Output
1	10
2	11
3	12
4	13

- a. -9 b. $\times 10$
c. $\times 4$ d. +9

10 Position → Value — which rule?

Position	Value
1	33
2	34
3	35
4	36

- a. $\times 33$ b. -32
c. $\div 33$ d. +32

Tip — An **Input/Output table** is a mini computer! Check the rule on **two** rows before you trust it — the **same rule must work for every row**. That's exactly how Scratch's **green Operator blocks** will behave in Chapter 7.



Logical Reasoning · Practice Sheet 3

HOMEWORK

CHAPTER 2
VEDIC WARM-
UP

- 1 What comes next in the sequence 2, 5, 8, 11
 - a. 13
 - b. 14
 - c. 15
 - d. 12
- 2 Observe the pattern: Triangle, Square, Circle, Triangle, Square, Circle, Triangle, Square, .What is the next shape?
 - a. Square
 - b. Triangle
 - c. Circle
 - d. Rectangle
- 3 If today is Tuesday, and 3 days ago it was Saturday, what day will it be 5 days from today?
 - a. Sunday
 - b. Monday
 - c. Tuesday
 - d. Wednesday
- 4 Complete the number pattern: 1, 4, 9, 16,...
 - a. 20
 - b. 25
 - c. 30
 - d. 22
- 5 A clock shows 3:00 PM. If the minute hand moves 180 degrees clockwise, what time will it be?
 - a. 3:30 PM
 - b. 9:00 PM
 - c. 3:15 PM
 - d. 6:00 PM
- 6 Which number comes next in the series: 100, 90, 81, 73...?
 - a. 66
 - b. 64
 - c. 65
 - d. 68
- 7 A plant grows cm on Monday, cm on Tuesday, cm on Wednesday, and so on. How many centimeters will it grow on Friday?
 - a. 5 cm
 - b. 6 cm
 - c. 7 cm
 - d. 8 cm
- 8 Identify the missing letter in the sequence: A, C, E, G,..., K?
 - a. I
 - b. H
 - c. J
 - d. M
- 9 A baker bakes cakes on Day 1, cakes on Day 2, cakes on Day 3. If this pattern continues, how many cakes will he bake on Day 5?
 - a. 6
 - b. 4
 - c. 2
 - d. 0
- 10 What is the next arrangement in the pattern?(Assume the arrow rotates clockwise by 90 degrees each time).Arrow pointing Up, Arrow pointing Right, Arrow pointing Down, .
 - a. Left
 - b. Up
 - c. Down
 - d. Right

CHAPTER FOUR

03 Loops

*Why write the same step 100 times when you can tell the computer
"do this 100 times" just once?*

REPEAT

FOREVER

EKADHIKENA PURVENA

DRAW A SQUARE

EXERCISE



3.1 Why do we need Loops?

CHAPTER 3

LOOPS



Big Idea

A **loop** tells the computer to **repeat** a set of steps. Loops save time, save typing, and let one tiny piece of code do BIG things.

Loops in real life

Brushing

Move brush up-down, up-down, up-down... 40 times. That's a loop!

Climbing stairs

Lift foot → step up → repeat for every step till you reach the top.

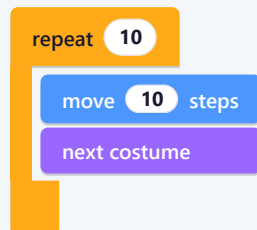
Breathing

Breathe in, breathe out — **forever**. Your body never stops looping.

Two famous loop blocks in Scratch

repeat (a fixed number of times)

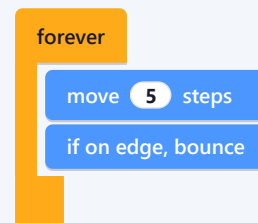
REPEATS INSIDE BLOCKS 10 TIMES, THEN STOPS



Use this when you know **exactly how many** times you want something to happen.

forever (never stops)

KEEPS RUNNING UNTIL YOU PRESS THE RED STOP BUTTON



Use this for things that should **keep going** — like a sprite that always follows the mouse.

The Magic Bridge · Vedic Sutra meets Code

VEDIC SUTRA

Ekadhikena Purvena

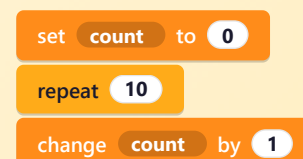
"By one more than the previous."

Counting **1, 2, 3, 4...** is nothing but starting at 1 and adding **one more** every time. That is exactly what a loop does in code.

IN SCRATCH

repeat & counter

Each time the loop runs, our counter becomes **one more than the previous** — the sutra in action!





3.2 Vedic Warm-up · Ekadhikena

CHAPTER 3
VEDIC WARM-UP



Big Idea · "One more than the previous"

The Vedic sutra **Ekādhikena Pūrveṇa** means "*by one more than the previous*". When we mark a digit with a little star ★, the rule is simple: **add 1 to that digit only**. Every other digit stays exactly the same.

RULE OF THE STAR ★

Ekadhik = add 1 to the marked digit.

This tiny rule powers many Vedic maths tricks — squaring numbers ending in 5, base-10 arithmetic, and (as we'll soon see) the way loops **count up** inside a computer.

How to solve — in 3 easy steps

1 Spot the star ★

Read the question — e.g. "*Of 6 in 65*" — and find the digit it points to. Draw a tiny star above it: **6★5**.

2 Add 1 to it

Only the marked digit changes. **6 → 7, 8 → 9, 2 → 3**. Everything else is untouched.

3 Rewrite the number

Put the new digit back in the same place. Copy the other digits exactly. That's your answer!

Worked examples

Ekadhikena · walk-through

Ekādhikena Pūrveṇa · "by one more than the previous"

OF 4 IN 4

Only one digit — mark it: **4★**. Add 1 → **5**

OF 8 IN 8

Only one digit — mark it: **8★**. Add 1 → **9**

OF 2 IN 42

Mark the 2: **4 2★**. The 4 stays; the 2 becomes 3 → **43**

Your turn · Practice sheet

A ★ next to a digit means "*this is the marked one*". Add 1 to that digit, keep the rest the same, and write your answer in the last column.

SR.	QUESTION	EKADHIK	ANSWER	SR.	QUESTION	EKADHIK	ANSWER
1	Of 6 in 65			7	Of 1 in 101		
2	Of 5 in 57			8	Of 5 in 35		
3	Of 9 in 91			9	Of 4 in 248		
4	Of 3 in 73			10	Of 6 in 462		
5	Of 7 in 127			11	Of 2 in 823		
6	Of 8 in 286			12	Of 9 in 192		

BRIDGE TO SCRATCH

Every time a Scratch **loop** runs, its counter jumps **one more than before** — $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \dots$ That is Ekadhiken at work inside a computer! Nail this warm-up and the **Draw a Square** project coming up will feel like a game you already know.



3.3 Vedic Warm-up · Ekadhiken Purvena

CHAPTER 3
VEDIC WARM-UP



Big Idea · "One more than the *previous*"

Purvena (also spelt *Pūrva* / *Purvna*) is the Sanskrit word for "**previous**" — the digit that sits **just to the left** of the one we're looking at. So **Ekadhiken Purvena** means: **add 1 to the digit that comes before the marked digit**. The marked digit itself does not change!

RULE OF THE LEFT-NEIGHBOUR

Ekadhiken Purvena = add 1 to the digit on the LEFT of the marked digit.

The marked digit and all the others stay exactly the same. *Different* from plain Ekadhiken — here it's the **left neighbour** that gains a 1, not the marked digit.

How to solve — in 3 easy steps

1 Star the marked digit



Read the question — e.g. "*Purvena of 7 in 378*" — and mark the digit it names: **3 7★ 8**.

2 Look one step to the

LEFT ←

Find the digit sitting **just before** the star. That is the **Purvena** (previous) digit. In **3 7★ 8**, the previous is **3**.

3 Add 1 to *that* digit

Bump the previous digit up by 1. Keep every other digit — including the marked one — exactly as it was. **3 → 4**, so **378** becomes **478**.

Worked examples

Ekadhiken Purvena · walk-through

Purvena · "of the previous"

**PURVENA OF 7 IN
378**

Mark the 7: **3 7★ 8**. The previous (left) is **3**. $3+1=4 \rightarrow$ **478**

**PURVENA OF 8 IN
684**

Mark the 8: **6 8★ 4**. The previous is **6**. $6+1=7 \rightarrow$ **784**

**PURVENA OF 9 IN
590**

Mark the 9: **5 9★ 0**. The previous is **5**. $5+1=6 \rightarrow$ **690**

Your turn · Practice sheet

Star the marked digit, look one step to its **LEFT**, and add 1 to that neighbour. Write the new number in the last column.

*Tip: if the marked digit has **no** neighbour on the left, imagine a tiny **0** sitting there — then add 1 to that 0.*

SR.	QUESTION	EKADHIK	ANSWER	SR.	QUESTION	EKADHIK	ANSWER
1	Purvena of 5 in 54			8	Purvena of 4 in 947		
2	Purvena of 7 in 378			9	Purvena of 1 in 213		
3	Purvena of 8 in 684			10	Purvena of 7 in 1705		
4	Purvena of 3 in 931			11	Purvena of 5 in 451		
5	Purvena of 6 in 4628			12	Purvena of 8 in 9876		
6	Purvena of 9 in 590			13	Purvena of 3 in 330		
7	Purvena of 2 in 120			14	Purvena of 6 in 961		
				15	Purvena of 2 in 2045		

EKADHIKEN VS. EKADHIKEN PURVENA

Ekadhiken adds 1 to the **marked** digit itself. **Ekadhiken Purvena** adds 1 to the digit **just before** it. Same "+1", different target! You'll see the **Purvena** rule return in Chapter 7 when we **square any number ending in 5** — that trick is Ekadhiken Purvena in disguise.



3.4 Mini-Project · Draw a Square

CHAPTER 3

LOOPS

HOMEWORK

TAKE-HOME PROJECT

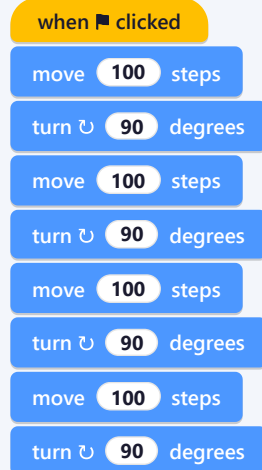
A square has **4 equal sides** and **4 right angles**. To draw it, our sprite must do the same two steps four times: **move** forward, then **turn** 90°. Perfect job for a loop!

Step 1 · Set up the pen

Scratch has a built-in **Pen** extension that lets a sprite draw on the stage. Click the blue **"Add Extension"** button at the bottom-left of the screen and choose **Pen**. New dark-green blocks will appear!

Without a loop 🙄

8 BLOCKS. SO MUCH TYPING!



With a loop 😁

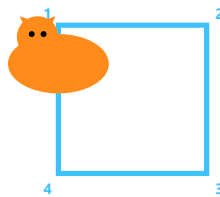
JUST 4 BLOCKS — SAME RESULT!



"pen down" is from the Pen extension — it acts like a coloured marker stuck to the sprite.

What you'll see on the Stage 🎬

Stage



repeat 4 times
→ move 100 → turn 90

The sprite draws all four sides of a perfect square, one corner after another.

Think like a coder 🗨️

What if we change repeat to 3?

You will get a **triangle**... *almost*! The turn must change to **120°**, because $360 \div 3 = 120$.

What if we change repeat to 6?

You'll draw a **hexagon** — use a **60°** turn. ($360 \div 6 = 60$). See the pattern?

VEDIC CONNECT

One more than the previous

From triangle (3 sides) → square (4) → pentagon (5)
→ hexagon (6)... every new shape is **Ekadhikena**
Purvena — one more side than the previous!

TRY THIS

Shape factory

→ Change **repeat** **N** and turn by **$360 \div N$** degrees.
Make a new shape every minute and stick them in
your workbook!

YOUR TURN · ACTIVITY 4.A

Build the Shape Factory

1. Make the cat draw a **square** (repeat 4, turn 90).
2. Stop, change to repeat 3 + turn 120 — get a **triangle**.
3. Now try repeat 6 + turn 60 — get a **hexagon**.
4. Bonus 🌟: repeat 36 + turn 10 — what shape did you get? Draw your guess below:



ASK A.I. Stuck? Try this prompt —

In Scratch, my cat draws a **square** using **repeat 4** → **move 100** → **turn 90°** with the Pen extension. I want it to draw a **hexagon (6 sides)** instead. What number should I put inside **repeat** ___ and **turn** ___ **degrees**? Please share the formula so I can try other shapes too (octagon, decagon).

💡 Asking for the **formula** is smarter than asking for one answer — you learn the rule, not just the result.



3.5 Exercises · Loops

CHAPTER 3 PRACTICE



Repeat & Forever — your turn!

Save your fingers from doing the same thing 100 times. Let the loop do it for you.

1

COUNT IT

How many times?

Each row shows a loop block. Write how many times the inside blocks will run.

Loop	Inside runs how many times?
repeat 10	
repeat 25	
forever	_____ until you press ■
repeat 4 inside repeat 3 (loops <i>inside</i> loops)	

2

MATCH THE SHAPE

repeat × turn = ?

Fill in the missing pieces. **Rule:** turn = $360 \div \text{number of sides}$.

Shape	repeat	turn (degrees)
Triangle	3	
Square		90°
Pentagon	5	
Octagon		

3

BUILD IT!

Rocket Countdown 🚀

Build a script that counts down from **10 to 1**, and then says "**BLAST OFF!**" for 2 seconds.

1. Make a variable called **counter** and set it to 10.
2. Use a **repeat 10** loop — inside it, the cat says **counter**, waits 1 sec, then **changes counter by -1**.
3. After the loop, the cat shouts "**BLAST OFF!**"

Which value should you put inside **change counter by** ___ ?

☐ My rocket counted down successfully!

Tip — A loop is just "*do this again*". If you find yourself copy-pasting blocks more than twice — use a loop.



Vedic Addition · Practice

CHAPTER 6

PRACTICE



Solve using your favourite Vedic trick!

Use **Make Tens**, the **Dot Method**, or **Left-to-Right**. Write the sum on the answer line under each stack.

1.

$$\begin{array}{r} 4 \\ 7 \\ + \quad 5 \\ \hline \end{array}$$

2.

$$\begin{array}{r} 8 \\ 3 \\ + \quad 6 \\ \hline \end{array}$$

3.

$$\begin{array}{r} 9 \\ 5 \\ + \quad 2 \\ \hline \end{array}$$

4.

$$\begin{array}{r} 6 \\ 8 \\ + \quad 4 \\ \hline \end{array}$$

5.

$$\begin{array}{r} 7 \\ 9 \\ + \quad 3 \\ \hline \end{array}$$

6.

$$\begin{array}{r} 5 \\ 4 \\ + \quad 8 \\ \hline \end{array}$$

7.

$$\begin{array}{r} 3 \\ 7 \\ + \quad 9 \\ \hline \end{array}$$

8.

$$\begin{array}{r} 2 \\ 8 \\ + \quad 6 \\ \hline \end{array}$$

9.

$$\begin{array}{r} 9 \\ 4 \\ + \quad 5 \\ \hline \end{array}$$

10.

$$\begin{array}{r} 8 \\ 6 \\ + \quad 7 \\ \hline \end{array}$$

11.

$$\begin{array}{r} 348 \\ 521 \\ + \quad 146 \\ \hline \end{array}$$

12.

$$\begin{array}{r} 672 \\ 184 \\ + \quad 239 \\ \hline \end{array}$$

13.

$$\begin{array}{r} 405 \\ 398 \\ + \quad 507 \\ \hline \end{array}$$

14.

$$\begin{array}{r} 918 \\ 276 \\ + \quad 134 \\ \hline \end{array}$$

15.

$$\begin{array}{r} 731 \\ 529 \\ + \quad 168 \\ \hline \end{array}$$

16.

$$\begin{array}{r} 892 \\ 234 \\ + \quad 672 \\ \hline \end{array}$$

Tip — For single-digit stacks (Q1–10), first scan for pairs that make **10**. For 3-digit stacks (Q11–15), use the **Dot Method** — one column at a time!

- 1** Consider the number sequence: 5, 9, 13, 17, If this pattern continues, what will be the 10th number in the sequence?

a. 37
b. 41
c. 45
d. 39

2 A robot is programmed to move 3 steps forward, then turn right, then move 3 steps forward, then turn left. This sequence is repeated. If the robot starts facing North, what direction will it be facing after completing the sequence 5 times?

a. North
b. East
c. South
d. West

3 A clock strikes once at 1 o'clock, twice at 2 o'clock, and so on. How many times will the clock strike in a 12-hour cycle from 1 o'clock to 12 o'clock?

a. 66
b. 78
c. 72
d. 12

4 A factory produces 25 toys every hour. If the factory operates for 8 hours a day, and takes a 1-hour break after every 3 hours of production, how many toys will be produced in a day?

a. 150
b. 175
c. 180
d. 180

5 Observe the pattern: **A, B, C, D, A, B, C, D,** What will be the 25th letter in this sequence?

a. A
b. B
c. C
d. D

6 A snail climbs 8 cm up a wall every hour, but slides down 3 cm during the next hour. If the wall is 25 cm high, how many hours will it take for the snail to reach the top?

a. 5 hours
b. 6 hours
c. 7 hours
d. 8 hours

7 A shop sells chocolates in packs of 6. If you buy a pack, you get 2 extra chocolates free. How many chocolates will you have if you buy 5 packs?

a. 30
b. 40
c. 35
d. 32

8 The numbers in a sequence follow the rule: start with 1, then multiply by 2 and add 1 to get the next number. What is the 4th number in this sequence?

a. 7
b. 15
c. 31
d. 11

9 A gardener plants a tree every 2 meters along a 50-meter path. If he plants a tree at the very beginning of the path, at 0 meters, how many trees will he plant in total?

a. 25
b. 26
c. 24
d. 23

10 A person exercises for 30 minutes every day. If they start exercising on a Monday, on which day of the week will they complete their 10th day of exercise?

a. Wednesday
b. Thursday
c. Friday
d. Saturday

11 A stack of coins has 3 coins in the first layer, 5 in the second, 7 in the third, and so on. If the pattern continues, how many coins will be in the 6th layer?

a. 11
b. 13
c. 15
d. 17

CHAPTER FOUR

04

Variables & Lists

Tiny labelled boxes inside the computer's brain — they remember your scores, names and answers so your code can do magic with them.

VARIABLES

SET / CHANGE

LISTS

EXERCISE



4.1 What is a Variable?

CHAPTER 4 VARIABLES & LISTS



Big Idea

A **variable** is a **labelled box** in the computer's memory. We give it a *name*, and we can put a *number* or a *word* inside. Later we can take it out, change it, or use it in our code.

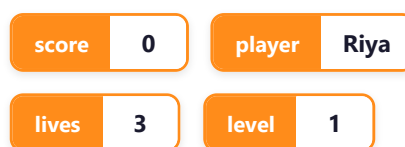
Think of variables like school lockers

Every locker has —

- A **name plate** ("Riya", "Aarav") so you know whose it is.
- An **inside** where you store something (a book, a tiffin).
- A **door** you can open to *change* what's inside.

A variable is exactly the same — only the locker is invisible and lives *inside* the computer.

My Scratch lockers right now —



The orange tag is the variable's **name**; the white box is the **value** inside.

How to make a variable in Scratch

1

Click the orange Variables category

It's the second-last category in the block palette.

2

Click "Make a Variable"

A small window pops up.

3

Type a name

Use a name that *describes* what's inside — like **score**, not **x**. Coders are clear, not lazy!

4

Click OK

Boom! ✨ New blocks appear and a tiny orange box pops up on the stage showing your variable.

The 4 super-power blocks of variables



Set

Put a *fresh* value into the box (throws away whatever was there before).

set **score** to **0**



Change

Add a number to whatever is already inside.

change **score** by **1**

 Show

Make the variable's box visible on the stage.

show variable **score**

 Hide

Hide the box from the stage (it still keeps the value!).

hide variable **score**

SET VS. CHANGE — CAREFUL!

set score to 5 means "make score equal to 5". **change score by 5** means "add 5 to whatever score already is".
Mixing them up is the #1 reason quiz scores look wrong!



4.2 Variables in Action · A Tiny Score

CHAPTER 4 VARIABLES & LISTS

Let's see how variables actually *change* while a program runs. Every line in the code below tells us what the variable looks like **at that moment**.

Walk-through · "Catch the Star" 🌟

The script

SPRITE: STAR · RUNS WHEN ■ CLICKED



What's inside *score* ?

Time	Block running	score
0 sec	flag clicked	score —
just after	set score to 0	score 0
1 sec	change by 1	score 1
2 sec	change by 1	score 2
3 sec	change by 1	score 3
...	forever	score ∞

Vedic Connect — variables are *Dhāraṇā* (मेमरी)

VEDIC IDEA

Hold the number in your head

In mental maths, you often **hold a partial answer** in your head while doing the next step (e.g. "2+3=5... now add 7... now 12"). That holding place is a variable!

IN SCRATCH

The variable IS your scratchpad

Use a variable called **running_total** or **answer** to remember the in-between number. Your sprite never forgets.

YOUR TURN · ACTIVITY 5.A

Coin Counter

1. Make a variable called **coins**.
2. Write a script: **when ■ clicked** → **set coins to 0**.
3. Add another script: **when this sprite clicked** → **change coins by 1**.
4. Click your sprite 5 times. The coins box on the stage should jump from 0 → 5. ✓

☐ I made my first counter!



4.3 What is a List?

CHAPTER 4 VARIABLES & LISTS



Big Idea

A variable holds **one** thing. A **list** is many variables in a neat row — each item has a number called its **index**, starting from **1**.

Lists you already use every day



Shopping list

Bread, milk, eggs, apples — items listed in order so nothing is forgotten.



Class timetable

Period 1, Period 2, Period 3... each slot has a subject.



Top scorers

1st, 2nd, 3rd place — exactly like a list of scores in a game.

A list looks like this on the stage

List name: *quiz_answers*

quiz_answers	length 5
1	99
2	88
3	9801
4	12
5	144
+	

Position 1 holds 99, position 2 holds 88, and so on.

The 5 most useful list blocks

add **5** to **quiz_answers**

delete **1** of **quiz_answers**

delete all of **quiz_answers**

replace item **2** of **quiz_answers** with **42**

item **3** of **quiz_answers**

Use **add** when collecting answers as the player plays. Use **item N of ...** to read an answer back later.

Building a list as the user plays

A LIST GROWS AUTOMATICALLY AS THE PLAYER ANSWERS EACH QUESTION

when **clicked**

delete all of **quiz_answers**

repeat **5**

ask **What is the answer?** and wait

add **answer** to **quiz_answers**

INDEX STARTS AT 1

Most professional coding languages start counting at 0 — but **Scratch is friendly** and starts at 1, just like a class register.



4.4 Mini-Project · Score Keeper

HOMEWORK

CHAPTER 4
VARIABLES &
LISTS



What we're building

A 5-question mental-maths quiz. The cat **asks** a question, the player **types** the answer, and Scratch keeps the **score** in a variable + saves every reply in a list. (We'll add the "Right or Wrong" check in the next chapter — for now, just the structure!)

Variables & List we'll need



score (variable)

Starts at 0. Goes up by 1 each correct answer.



q_no (variable)

Tells us which question we're on (1, 2, 3, 4, 5).



player_answers (list)

Stores every answer the player typed.

The complete script

SPRITE: MATHS-CAT · RUNS WHEN ■ CLICKED

```
when clicked
  set score to 0
  set q_no to 0
  delete all of player_answers
  say Welcome! 5 quick Vedic questions. for 2 secs
  repeat 5
    change q_no by 1
    ask Question... and wait
    add answer to player_answers
  say Quiz over! Check your answers list for 3 secs
```

After the quiz, the stage looks like —

score0

q_no5

Score is still 0 because we haven't checked answers yet (that's Chapter 6's job). But **q_no** proudly shows we asked all 5 questions.

player_answers		length 5
1	99	
2	88	
3	9801	
4	12	
5	144	

YOUR TURN · ACTIVITY 5.B

Type your 5 Vedic questions

Plan your quiz *before* coding. Write 5 fun mental-maths questions you'd like the cat to ask:

1. Q1: _____
2. Q2: _____
3. Q3: _____
4. Q4: _____
5. Q5: _____

☐ I built the quiz skeleton!



ASK A.I.

Stuck? Try this prompt —

I'm building a Scratch quiz with a variable called **score** and a list called **player_answers**. After each question I want to **add the player's answer to the list** and **show the score** on the stage. But every time the cat asks the next question, my score box shows **0**. I'm using **set score to 0** at the start. What might be making the score reset, and how do I fix it?

💡 Telling A.I. the **variable names** you used (*score, player_answers*) helps it explain in the same language as your code.



4.5 Exercises · Variables & Lists

CHAPTER 4 PRACTICE



Boxes that remember!

Track values, fix bugs and build your own little memory machine.

1

TRACE IT

What's in the score box?

Start: **score** = 0. Fill the table after each instruction.

Instruction	score is now...
change score by 5	
change score by 3	
set score to 10	
change score by -4	
change score by 1	

2

LIST TRACE

What's in the list?

Start with an **empty** list called **fruits**. Apply the steps in order, then write the final list:

1. add **"apple"** to fruits
2. add **"banana"** to fruits
3. add **"cherry"** to fruits
4. delete item **1** of fruits
5. add **"date"** to fruits

Final fruits list (in order):

1

2

3

4

3

BUILD IT!

Friend Collector 👤👤

Build a project that asks the player for **3 friends' names** and stores them in a list called **friends**. Then make the cat say each name one by one.

- Use the **ask ... and wait** block (Sensing) to get a name.
- Use **add (answer) to friends** to save it.
- Use a **repeat 3** loop with **item (i) of friends** to read them back.

☐ My cat remembers all 3 friends!

Tip — A **variable** holds *one* value. A **list** holds *many* values. Same idea, different sizes.

4.5 Worksheet • Logical Reasoning

CHAPTER 4

WORKSHEET

Variable Logic Challenge

Solve this worksheet to strengthen algebra thinking, missing-number patterns and list ordering skills.

- 1** If $A + 12 = 30$, what is the value of A?
a. 18
b. 42
c. 20
d. 12

2 A shopkeeper had X balloons. He sold 15 balloons and now has 23 balloons left. How many balloons did he have initially?
a. 8
b. 38
c. 15
d. 23

3 What is the missing number in the sequence: 5, 10, 15, _, 25, 30?
a. 18
b. 20
c. 22
d. 24

4 If $4 \times Y = 36$, what is the value of Y?
a. 8
b. 7
c. 9
d. 12

5 Find the next number in the pattern: 70, 63, 56, 49, _?
a. 42
b. 40
c. 35
d. 50

6 Sarah shares N candies equally among 6 friends. Each friend gets 7 candies. How many candies did Sarah have initially?
a. 13
b. 1
c. 42
d. 36

7 A list of numbers is given: 25, 18, 32, 11, 40. If these numbers are arranged from smallest to largest, which number will be in the middle?
a. 18
b. 35
c. 32
d. 11

8 What comes next in the series: 2, 3, 5, 8, 12, _?
a. 15
b. 17
c. 18
d. 20

9 If $P + 5 = 13$ and $Q - 3 = 6$, what is the value of $P + Q$?
a. 8
b. 9
c. 17
d. 10

10 Which number does not belong to the list: 10, 20, 30, 35, 40?
a. 10
b. 20
c. 35
d. 40

- 11** An animal has **L** legs. If there are 3 animals and a total of 12 legs, how many legs does one animal have?

a. 3 b. 4
c. 6 d. 12

12 A basket contains **R** red apples and **G** green apples. There are 7 red apples. If the total number of apples is 15, how many green apples are there?

a. 7 b. 8
c. 15 d. 22

13 What is the missing number in the sequence: **3, 6, 12, _, 48, 96**?

a. 18 b. 20
c. 24 d. 36

14 A number **N** is greater than 10 but less than 15. It is an odd number. What is N?

a. 11 b. 12
c. 14 d. 10

15 If **Square = 6** and **Triangle = 4**, what is the value of **Square + Triangle - 3**?

a. 7 b. 10
c. 13 d. 11

16 In the list of numbers: **8, 13, 5, 20, 10**. If you arrange them in ascending order, what is the third number?

a. 8 b. 10
c. 13 d. 5

17 There are 4 rows of chairs in a classroom. Each row has **C** chairs. If there are a total of 24 chairs, how many chairs are in each row?

a. 4 b. 6
c. 8 d. 20

18 What is the next number in the series: **1, 4, 2, 5, 3, 6, _**?

a. 4 b. 7
c. 8 d. 9

19 A square has a side length of **S** cm. Its perimeter is 28 cm. What is the length of one side?

a. 4 cm b. 7 cm
c. 14 cm d. 28 cm

20 Raj has 12 toy cars. He gives **X** cars to his friend and now has 7 cars left. How many cars did Raj give to his friend?

a. 5 b. 7
c. 12 d. 19

CHAPTER FIVE

05

Conditionals (if / else)

Teach your sprite to think — to ask "is this true?" and choose what to do next, just like you do every day.

IF

IF / ELSE

COMPARISONS

QUIZ CHECKER



5.1 Yes or No · The Condition

CHAPTER 5 CONDITIONALS



Big Idea

A **condition** is a question that has only two answers — **YES** (true) or **NO** (false). Code uses conditions to **choose** what happens next.

You make condition-decisions every day



Is it raining?

YES → take an umbrella.
NO → leave it at home.



Is the milk hot?

YES → wait 5 minutes.
NO → drink it now.



Did I finish my homework?

YES → play.
NO → finish first!

In code, this looks like a fork in the road

IF YES

Run the green blocks — go this way.

answer = 99 ?

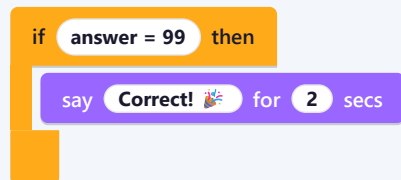
IF NO

Run the red blocks — go that way instead.

The two control blocks you'll use

if < ... > then

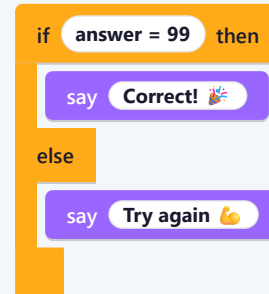
RUN THE INSIDE BLOCKS ONLY WHEN ANSWER IS YES



Use this when there's **only something to do if YES** — and nothing to do if NO.

if < ... > then / else

TWO PATHS — ONE FOR YES, ONE FOR NO



Use this when **both** answers (YES and NO) need to do something.



5.2 Asking Yes/No · the green Operator blocks

CHAPTER 5 CONDITIONALS

Inside the diamond hole at the top of an *if* block, you slot a **green operator block**. That block asks the YES/NO question — and Scratch answers it for you.

The 3 most-used comparison blocks

= Equal to

5 = 5

YES if both sides are exactly the same.

< Less than

3 < 7

YES if the *left* number is smaller.

> Greater than

9 > 2

YES if the *left* number is bigger.

Mini-quiz · what's the answer?

Read each green block and circle YES or NO

10 = 10

YES NO

5 < 3

YES NO

7 > 2

YES NO

99 = 100

YES NO

Combining conditions

Sometimes one question isn't enough. You can **combine** two questions using:

☐ and ☐

☐ or ☐

not ☐

and = both must be YES. **or** = at least one is YES. **not** = flips YES↔NO.



The Hub for Young Next-gen Knowledge

Real-life example · "Can I go play?"

TWO CONDITIONS JOINED WITH "AND" — BOTH MUST BE YES

if homework_done = yes and time < 7 then

say Yay! Time to play 🎮

else

say Finish work first 📅

VEDIC CONNECT

IN SCRATCH



Yāvadūnam — "*by the deficiency*"

Vedic Maths often asks: "How far is this number **from 100**?" — that's a comparison question, just like our = and < blocks.

Compare → decide → act

The green operator does the comparing. The orange **if** block decides. Together they make your sprite *think*.



5.3 Mini-Project · Quiz Checker

CHAPTER 5
CONDITIONALS

HOMEWORK

TAKE-HOME PROJECT



What we're upgrading

We're going back to the Vedic Quiz from Chapter 5 and giving the cat a **brain**. After every question, the cat will *compare* the player's answer with the correct Vedic answer and say "**Right!**" or "**Try again**".

The plan

1

Ask the question

Use the blue **ask** block. The player's reply lands in **answer** automatically.

2

Compare with the correct answer

Use the green **=** block: *answer = 99 ?*

3

Use if-else to give feedback

YES → say "Right!" + change score by 1. NO → say "Almost — the answer was 99".

4

Repeat for all 5 questions

(Or use a list of correct answers — that's a Grade-5 trick!)

Sample question · 99×1 (using *Nikhilam* sutra)

The script for ONE question

ADD THIS INSIDE THE REPEAT-5 LOOP

```
ask 100 - 1 = ? and wait
if answer = 99 then
  say Right! 🎉 for 1 sec
  change score by 1
else
  say Oops! It was 99 for 2 secs
```

What you'll see on the Stage 🎬

Stage

score 3



Right! 🎉

100 - 1 = ?

99

When the player answers 99, the cat cheers and the score box ticks up.

VEDIC SUTRA

IN SCRATCH

Navataścaramaṁ Daśataḥ

"All from 9 and the last from 10." — the trick we used to compute $100 - 1 = 99$ in one second, no borrowing needed.

**The if-else is the judge**

The cat doesn't know maths — it just **compares** the player's number with the correct one. Smart code, simple thinking.

YOUR TURN · ACTIVITY 6.A**Add 4 more checked questions**

1. Copy the if-else above 4 more times — once for each of your 5 questions.
2. Change the question text and the correct answer in the green = block.
3. Test by purposely typing *wrong* answers — the cat should say "Oops!" 🙊

☐ My quiz now scores by itself!

**ASK A.I. Stuck? Try this prompt —**

In my Scratch quiz the cat asks "What is $100 - 1$?". I want it to say "Right!" if the player's `answer = 99`, and "Oops, try again!" otherwise. I'm using the `if ... else` control block with the green = operator inside, but the `else` part never runs even when I type a wrong number. Can you check what might be wrong with my comparison?

💡 Naming the **blocks you used** (if...else, green =) helps A.I. point straight at the bug.



Exercises · Conditionals

CHAPTER 5
PRACTICE



If... then... else!

Make the computer think. Decide. Choose. Just like you do every day.

1 TRUE OR FALSE? Tick the right box.

Condition	True	False
10 > 5	☐	☐
7 = 7	☐	☐
"cat" = "Cat"	☐	☐
100 < 99	☐	☐
(3 + 2) = 5	☐	☐

2 PREDICT What does the cat say?

The cat asks: "Pick a number." For each input below, write what the cat replies.

```
when clicked
ask Pick a number and wait
if answer > 10 then
  say Big!
else
  say Small!
```

If you type...	Cat says...
15	
10	
3	
100	

3 BUILD IT! Pass / Fail Reporter

Build a script that asks the player for their marks (out of 100) and replies:

- If marks ≥ 90 → say "Excellent! 🌟 "
- Else if marks ≥ 50 → say "Pass 😊 "
- Else → say "Try again 🙌 "

How many if ... else blocks do you need (nested)?

☐ I built a 3-level grader!

Tip — Conditions are just questions with yes or no answers. The cat acts only when the answer is yes (true).



5.5 Worksheet · If / Else Logic

CHAPTER 5 WORKSHEET



Condition Logic Challenge

Solve this worksheet to practise if/else decisions, AND/OR logic and branch-by-branch reasoning.

- 1 If a number is greater than 50, add 10 to it. Otherwise, if it is 50 or less, subtract 5 from it. What is the result for the number 55?

a. 65	b. 50
c. 60	d. 45
- 2 A fruit is considered "ripe" if its color is red AND its size is large. If its color is red OR its size is large, but not both, it is "almost ripe". Otherwise, it is "unripe". A fruit is red and small. What is its status?

a. Ripe	b. Almost ripe
c. Unripe	d. Overripe
- 3 A student passes an exam if their score is 75 or more. If they score between 50 and 74 inclusive, they need to re-appear. If they score less than 50, they fail. A student scores 68. What is the outcome?

a. Pass	b. Fail
c. Re-appear	d. Excellent
- 4 If today is Monday, then tomorrow is Tuesday. Otherwise, if today is Saturday, then tomorrow is Sunday. In all other cases, tomorrow is a weekday. If today is Wednesday, what is tomorrow?

a. Tuesday	b. Sunday
c. Thursday	d. Monday
- 5 A toy costs **Rs. 120**. If a customer buys 2 or more toys, they get a **Rs. 10** discount per toy. If they buy less than 2 toys, they get no discount. How much will a customer pay if they buy 3 toys?

a. Rs. 360	b. Rs. 350
c. Rs. 330	d. Rs. 390
- 6 Consider a number. If it is an even number AND it is less than 20, then multiply it by 2. Otherwise, if it is an odd number AND greater than 10, then add 5 to it. In all other scenarios, subtract 1. What is the result for the number 16?

a. 32	b. 15
c. 21	d. 17
- 7 A child gets extra playtime if they finish their homework AND clean their room. If they finish only one of these tasks, they get regular playtime. If they do neither, they get no playtime. The child finished homework but did not clean the room. What kind of playtime do they get?

a. Extra playtime	b. Regular playtime
c. No playtime	d. Half playtime
- 8 If an animal has feathers, it is a bird. If it has fur AND lays eggs, it is an echidna. Otherwise, if it has fur AND gives birth to live young, it is a mammal. What kind of animal is one that has fur and gives birth to live young?

a. Bird	b. Echidna
c. Mammal	d. Reptile
- 9 A bookstore offers special prices. If a book costs more than **Rs. 500**, its price is reduced by **Rs. 100**. If a book costs between **Rs. 200** and **Rs. 500**, inclusive of Rs. 200 but not Rs. 500, its price is reduced by **Rs. 50**. Otherwise, the price remains the same. What is the final price of a book that originally costs **Rs. 450**?

a. Rs. 350	b. Rs. 400
c. Rs. 450	d. Rs. 550

PRACTICE PLAN**How to use this worksheet**

1. Underline the condition words first: if, otherwise, AND and OR.
2. For each question, follow only the branch that matches the situation.
3. Check the answer key after solving, then rewrite two tricky rules as Scratch if/else blocks.

CHAPTER SIX

06

Operators & Maths Blocks

The green family — the maths brain of Scratch. We'll bring three famous Vedic sutras to life inside our code.

+ - × ÷

JOIN & PICK

VEDIC ADDITION

VEDIC SUBTRACTION

NIKHILAM

URDHVA-TIRYAGBHYĀM



6.1 The Green Operators Family

CHAPTER 6 OPERATORS & MATHS



Big Idea

Operators are the **maths and logic blocks**. They are shaped like **green pills** because they always *fit inside* other blocks — they don't run on their own, they help other blocks decide and calculate.

The maths operators (the four old friends)

+ Addition

5 + 3

Gives 8.

− Subtraction

9 − 4

Gives 5.

× Multiplication

6 × 7

Gives 42.

÷ Division

12 ÷ 4

Gives 3.

Two more very useful operators

pick random

pick random 1 to 10

Gives a surprise number every time. Perfect for quiz questions and dice.

join (glue text together)

join Score: score

Glues words and numbers into one sentence. Makes your sprite say smart things.

Operators slot *inside* other blocks

Notice how each green pill sits **inside** a white circle of another block. That's the magic of Scratch — small pieces fit together like LEGO!

3 WAYS OPERATORS ARE USED

set total to 5 + 3

say join Total = total for 2 secs

move pick random 10 to 50 steps

ORDER MATTERS IN MATHS TOO!

Scratch follows the same rules you learn in school: **BODMAS** (Brackets → Of → Division → Multiplication → Addition → Subtraction). Use **nested green pills** to control the order, e.g. $(2 + 3) \times 4 = 20$, but $2 + 3 \times 4 = 14$.



Vedic Addition · Practice

CHAPTER 6

PRACTICE



Level up — big numbers, same tricks!

Now try 3-digit and 4-digit sums. The **Dot Method** is your best friend here — one column at a time, top to bottom.

16.

$$\begin{array}{r} 482 \\ 316 \\ + \quad 295 \\ \hline \\ \hline \end{array}$$

17.

$$\begin{array}{r} 659 \\ 207 \\ + \quad 481 \\ \hline \\ \hline \end{array}$$

18.

$$\begin{array}{r} 874 \\ 125 \\ + \quad 699 \\ \hline \\ \hline \end{array}$$

19.

$$\begin{array}{r} 543 \\ 612 \\ + \quad 178 \\ \hline \\ \hline \end{array}$$

20.

$$\begin{array}{r} 296 \\ 483 \\ + \quad 721 \\ \hline \\ \hline \end{array}$$

21.

$$\begin{array}{r} 3482 \\ 5917 \\ 2604 \\ + \quad 1489 \\ \hline \\ \hline \end{array}$$

22.

$$\begin{array}{r} 7265 \\ 1834 \\ 5901 \\ + \quad 274 \\ \hline \\ \hline \end{array}$$

23.

$$\begin{array}{r} 9054 \\ 2178 \\ 3615 \\ + \quad 4203 \\ \hline \\ \hline \end{array}$$

24.

$$\begin{array}{r} 6429 \\ 1857 \\ 904 \\ + \quad 7136 \\ \hline \\ \hline \end{array}$$

25.

$$\begin{array}{r} 5298 \\ 3467 \\ 1859 \\ + \quad 6274 \\ \hline \\ \hline \end{array}$$

26.

$$\begin{array}{r} 8143 \\ 2765 \\ 3928 \\ + \quad 1456 \\ \hline \\ \hline \end{array}$$

27.

$$\begin{array}{r} 9731 \\ 2684 \\ 5419 \\ + \quad 3207 \\ \hline \\ \hline \end{array}$$

28.

$$\begin{array}{r} 4862 \\ 7135 \\ 2594 \\ + \quad 4081 \\ \hline \\ \hline \end{array}$$

29.

$$\begin{array}{r} 6509 \\ 1846 \\ 9723 \\ + \quad 3158 \\ \hline \\ \hline \end{array}$$

30.

$$\begin{array}{r} 4327 \\ 5896 \\ 2715 \\ + \quad 6048 \\ \hline \\ \hline \end{array}$$

☐ I finished all 30 — the Vedic way! ★



6.3 Vedic Algorithm 2 · Vedic Subtraction

CHAPTER 6

OPERATORS & MATHS



The Sutra

Nikhilam Navataścaramaṁ Daśataḥ — "All from 9 and the last from 10." This one sutra unlocks **super-fast subtraction**. With a little twist called the **Complement Trick**, we can subtract any number from any number — *without borrowing even once!*

Three Vedic super-tricks for subtraction

TRICK 1

Nikhilam · From 10, 100, 1000...

To find $1000 - 437$, apply the sutra: $9-4=5$, $9-3=6$, $10-7=3 \rightarrow 563$. Every digit from 9, the last one from 10. That's it!

TRICK 2

The Complement Trick

For any subtraction, **round the bottom number up** to the nearest 10, 100 or 1000. Add the tiny **complement** to the top, then subtract the round number. No borrowing, no scratching!

TRICK 3

Think "What Completes the Top?"

For small stacks like $15 - 7$, don't take away — *add up!* Ask: $7 + ? = 15$. Answer **8**. Your brain is already fast at addition; use that speed for subtraction too.

Worked example — the Complement Trick

Subtract: $452 - 76 = ?$ The bottom (76) is close to **100** — let's use that.

Complement Method · step by step		Nikhilam · complement to 100
STEP 1	Round the bottom (76) up to the nearest 100 $\rightarrow$ 100	
STEP 2	Find the complement of 76 using Nikhilam: $9-7=2$, $10-6=4 \rightarrow$ 24	
STEP 3	Add the complement to the top: $452 + 24 = 476$	
STEP 4	Subtract the round number: $476 - 100 = 376$	
ANSWER	376 $452 - 76 = 376$ (no borrowing!)	

Bigger example — 5604 - 2786

Same trick, bigger numbers. Round **2786** up to **3000**.

Steps	
5 6 0 4	1. Round 2786 up to 3000 .
	2. Complement = $3000 - 2786 = 214$.

$$\begin{array}{r} - 2786 \\ \hline \end{array}$$
$$2818$$
3. Add complement to top: $5604 + 214 = 5818$.4. Subtract the round number: $5818 - 3000 = 2818$ ✓*The bigger the numbers, the more time the trick saves!*

WHY VEDIC SUBTRACTION FEELS LIKE MAGIC

Regular subtraction makes you **borrow** across columns — messy and slow. The **complement trick** turns subtraction into *a small addition + a subtraction from a round number*. Both parts are easy. Try it on the worksheet — you'll never subtract the old way again!



Vedic Subtraction · Practice Part 1 · Q1–15

CHAPTER 6

PRACTICE



Warm-up — small numbers, big brains!

Start with the "What completes the top?" trick. From Q11 onwards, try the **Complement Trick** — round the bottom up.

1.

$$\begin{array}{r} 9 \\ - 4 \\ \hline \\ \hline \end{array}$$

2.

$$\begin{array}{r} 15 \\ - 7 \\ \hline \\ \hline \end{array}$$

3.

$$\begin{array}{r} 18 \\ - 9 \\ \hline \\ \hline \end{array}$$

4.

$$\begin{array}{r} 27 \\ - 8 \\ \hline \\ \hline \end{array}$$

5.

$$\begin{array}{r} 34 \\ - 6 \\ \hline \\ \hline \end{array}$$

6.

$$\begin{array}{r} 42 \\ - 9 \\ \hline \\ \hline \end{array}$$

7.

$$\begin{array}{r} 55 \\ - 7 \\ \hline \\ \hline \end{array}$$

8.

$$\begin{array}{r} 63 \\ - 8 \\ \hline \\ \hline \end{array}$$

9.

$$\begin{array}{r} 71 \\ - 9 \\ \hline \\ \hline \end{array}$$

10.

$$\begin{array}{r} 88 \\ - 5 \\ \hline \\ \hline \end{array}$$

11.

$$\begin{array}{r} 124 \\ - 8 \\ \hline \\ \hline \end{array}$$

12.

$$\begin{array}{r} 136 \\ - 19 \\ \hline \\ \hline \end{array}$$

13.

$$\begin{array}{r} 205 \\ - 17 \\ \hline \\ \hline \end{array}$$

14.

$$\begin{array}{r} 248 \\ - 39 \\ \hline \\ \hline \end{array}$$

15.

$$\begin{array}{r} 317 \\ - 28 \\ \hline \\ \hline \end{array}$$

16.

$$\begin{array}{r} 452 \\ - 76 \\ \hline \\ \hline \end{array}$$

Tip — For Q11–15, round the bottom up to the nearest **10**. Example: **136 – 19**. Round 19 → 20, complement = 1. So **136 + 1 – 20 = 137 – 20 = 117**. ✨



Vedic Subtraction · Practice Part 2 · Q16–30

CHAPTER 6 PRACTICE



Level up — the Complement Trick shines!

For each sum: round the **bottom** up to the nearest **100** or **1000**, add the tiny complement to the top, then subtract the round number.

17.

$$\begin{array}{r} 584 \\ - 98 \\ \hline \\ \hline \end{array}$$

18.

$$\begin{array}{r} 673 \\ - 157 \\ \hline \\ \hline \end{array}$$

19.

$$\begin{array}{r} 805 \\ - 286 \\ \hline \\ \hline \end{array}$$

20.

$$\begin{array}{r} 946 \\ - 458 \\ \hline \\ \hline \end{array}$$

21.

$$\begin{array}{r} 1234 \\ - 567 \\ \hline \\ \hline \end{array}$$

22.

$$\begin{array}{r} 2456 \\ - 789 \\ \hline \\ \hline \end{array}$$

23.

$$\begin{array}{r} 3810 \\ - 1945 \\ \hline \\ \hline \end{array}$$

24.

$$\begin{array}{r} 4728 \\ - 1859 \\ \hline \\ \hline \end{array}$$

25.

$$\begin{array}{r} 5604 \\ - 2786 \\ \hline \\ \hline \end{array}$$

26.

$$\begin{array}{r} 6842 \\ - 3957 \\ \hline \\ \hline \end{array}$$

27.

$$\begin{array}{r} 7420 \\ - 1865 \\ \hline \\ \hline \end{array}$$

28.

$$\begin{array}{r} 8531 \\ - 4978 \\ \hline \\ \hline \end{array}$$

29.

$$\begin{array}{r} 9164 \\ - 5879 \\ \hline \\ \hline \end{array}$$

30.

$$\begin{array}{r} 12035 \\ - 6847 \\ \hline \\ \hline \end{array}$$

31.

$$\begin{array}{r} 8989 \\ - 6767 \\ \hline \\ \hline \end{array}$$

32.

$$\begin{array}{r} 37832 \\ - 7816 \\ \hline \\ \hline \end{array}$$

☐ I finished all 32 subtractions — the Vedic way, no borrowing! ★



6.4 Vedic Algorithm 3 · Navataścaramaṁ Daśataḥ in Scratch



The Sutra

Nikhilam Navataścaramaṁ Daśataḥ — "All from 9 and the last from 10." A magical way to subtract from 100, 1000, 10000 — without borrowing!

How a human does it (mentally)

Goal: $1000 - 437 = ?$

Subtract from	1000	Nikhilam · "All from 9, last from 10"
STEP 1	4 → 9 - 4 = 5 (all from 9)	
STEP 2	3 → 9 - 3 = 6 (all from 9)	
STEP 3	7 → 10 - 7 = 3 (last from 10)	
ANSWER	5 6 3 ✓ $1000 - 437 = 563$	

Now let's teach Scratch to do it

We'll keep it simple — for any 3-digit number, our cat will **read each digit**, apply the sutra, and show the answer.

Variables we need

number437

d14

d23

d37

answer563

The script

STEP-BY-STEP NIKHILAM IN SCRATCH (FOR 1000 - NUMBER)

when clicked

askEnter a 3-digit number under 1000and wait

set number to answer

set d1 to letter 1 of number

set d2 to letter 2 of number

set d3 to letter 3 of number

set d1 to 9 - d1

set d2 to 9 - d2

set d3 to 10 - d3

set answer to join d1 join d2 d3

say join 1000 - number = answer for 3 secs

WHY IT WORKS

The **letter N of** operator (from the green family) lets us peek at one digit of a number — exactly what the sutra asks us to do, one digit at a time. We then **join** the three results to get the final answer.

YOUR TURN · ACTIVITY 7.A

Try it with three different numbers

Run your project and write your own results below.

Input	Cat said...	My check
1000 - 248	_____	_____
1000 - 591	_____	_____
1000 - 803	_____	_____
☐ My Scratch sprite knows Nikhilam!		



6.5 Vedic Algorithm 4 · Urdhva Multiplication

CHAPTER 6
OPERATORS &
MATHS



The Sutra

Urdhva-Tiryagbhyām — "Vertically and Crosswise." The most famous Vedic shortcut for multiplying two numbers in one line.

How a human does it (for two 2-digit numbers)

Goal: $23 \times 12 = ?$ Let's call it $(a\ b) \times (c\ d)$ where $a=2, b=3, c=1, d=2$.

Multiply	23×12	Urdhva · "Vertically & Crosswise"
VERTICAL (RIGHT)	$b \times d = 3 \times 2 = 6 \rightarrow$ ones place	
CROSSWISE	$(a \times d) + (b \times c) = (2 \times 2) + (3 \times 1) = 7 \rightarrow$ tens place	
VERTICAL (LEFT)	$a \times c = 2 \times 1 = 2 \rightarrow$ hundreds place	
ANSWER	2 7 6 ✓ $23 \times 12 = 276$ (carry handled if needed)	

In Scratch — the cat becomes a Vedic calculator

Variables we need

a	2	b	3	c	1	d	2	ones	6	tens	7
hundreds	2	result	276								

The script

READS TWO 2-DIGIT NUMBERS AND USES THE SUTRA TO MULTIPLY THEM

```

when clicked
ask First 2-digit number? and wait
set a to letter 1 of answer
set b to letter 2 of answer
ask Second 2-digit number? and wait
set c to letter 1 of answer
set d to letter 2 of answer
set ones to b × d
set tens to (a × d) + (b × c)

```

```

set hundreds to a X c
set result to hundreds X 100 + tens X 10 + ones
say join Answer = result for 3 secs
  
```

WHY THIS ROCKS**One line · One sutra · One answer**

The same sutra works for 3-digit × 3-digit, 4-digit × 4-digit, and beyond — just more crosswise pairs!

IN SCRATCH**Maths sutra → Maths code**

You just translated an ancient sutra into a working program. **That**, dear coder, is real magic. ✨

YOUR TURN · ACTIVITY 7.B**Vedic vs. Calculator Race**

1. Run your Scratch project and ask it: **23 × 12**, **14 × 21**, **32 × 13**.
2. Solve the same questions *mentally* using the sutra.
3. Time yourself with a stopwatch — who's faster, you or Scratch? 🕒

☐ I built a Vedic multiplier in Scratch!

**ASK A.I. Stuck? Try this prompt —**

I'm using the Urdhva-Tiryagbhyam (vertically & crosswise) sutra in Scratch to multiply two 2-digit numbers like **23 × 12**. I have variables **a, b, c, d, ones, tens, hundreds, result**. I'm stuck on the green **operator** blocks for **tens**, which should equal **(a × d) + (b × c)**. How do I **nest** the green × and + blocks correctly inside one **set tens to** block?

💡 The word "**nest**" tells A.I. you mean blocks-inside-blocks — super useful for the green Operators family.



6.6 Multiplication Warm-up 2-digit × 1-digit

CHAPTER 6 PRACTICE



Sharpen your multiplication muscle!

Warm up with 2-digit × 1-digit sums *before* we unlock the Vedic shortcuts on the next pages. Use **Urdhva-Tiryagbhyam** (vertically & crosswise) or the classic column method.

How to solve — the Vedic way (Urdhva applied to 2 × 1 digits)

Take a 2-digit number (**a b**) and a 1-digit number **c**. The answer is: (**a × c**) carried on to (**b × c**).

$$46 \times 28$$

Left
 4×2
 $\downarrow$
 8

Cross Multiply
 $4 \times 8 = 32$
 $+$
 $2 \times 6 = 12$
 $\downarrow$
 44

Right
 6×8
 $\downarrow$
 48

8

4 4

4 8

↓ Carry

1288

Exercise · Solve mentally, then write the answer

1.

$$\begin{array}{r} 12 \\ \times 16 \\ \hline \end{array}$$

2.

$$\begin{array}{r} 18 \\ \times 14 \\ \hline \end{array}$$

3.

$$\begin{array}{r} 19 \\ \times 13 \\ \hline \end{array}$$

4.

$$\begin{array}{r} 24 \\ \times 46 \\ \hline \end{array}$$

5.

$$\begin{array}{r} 27 \\ \times 11 \\ \hline \end{array}$$

6.

$$\begin{array}{r} 13 \\ \times 12 \\ \hline \end{array}$$

7.

$$\begin{array}{r} 14 \\ \times 38 \\ \hline \end{array}$$

8.

$$\begin{array}{r} 37 \\ \times 45 \\ \hline \end{array}$$

9.

$$\begin{array}{r} 56 \\ \times 10 \\ \hline \end{array}$$

10.

$$\begin{array}{r} 45 \\ \times 24 \\ \hline \end{array}$$

11.

$$\begin{array}{r} 97 \\ \times 21 \\ \hline \end{array}$$

12.

$$\begin{array}{r} 94 \\ \times 67 \\ \hline \end{array}$$

13.

$$\begin{array}{r} 75 \\ \times 36 \\ \hline \end{array}$$

4.

$$\begin{array}{r} 37 \\ \times 14 \\ \hline \end{array}$$

15.

$$\begin{array}{r} 40 \\ \times 20 \\ \hline \end{array}$$

16.

$$\begin{array}{r} 61 \\ \times 17 \\ \hline \end{array}$$

17.

6 9

5 1

x

18.

7 3

2 6

x

19.

1 9

2 9

x

20.

1 8

4 8

x

Tip — Try solving each one *without* writing intermediate steps. Vedic maths is about doing it in **one line, in your head**. Speed comes with practice — race yourself with a stopwatch!



6.7 Vedic Trick · Multiply by 11, 111 ... Part 1 · Q1–20

CHAPTER 6 PRACTICE

1 1

The "sum of neighbours" trick

Multiplying *any* number by **11** takes just one step in Vedic maths — no long multiplication needed. Learn the method, then blaze through the practice sums.

Describe the Method — $\times 11$ in one line

2-digit $A B \times 11$: write **A** (**A + B**) **B**. · **3-digit $A B C \times 11$:** write **A** (**A + B**) (**B + C**) **C**. · **4-digit $A B C D \times 11$:** write **A** (**A + B**) (**B + C**) (**C + D**) **D**. If any sum is ≥ 10 , keep the ones digit and *carry 1* to the left.

$27 \times 11 \rightarrow 2 _ 7$ with $(2+7)=9$
Answer = **297**

$78 \times 11 \rightarrow 7 _ 8$ with $(7+8)=15 \rightarrow$ carry
 $(7+1) _ 5 _ 8 =$ **858**

$134 \times 11 \rightarrow 1 (1+3) (3+4) 4$
= **1474**

$2843 \times 11 \rightarrow$ carry chain
= **31273**

1.

$$\begin{array}{r} 27 \\ \times 11 \\ \hline \end{array}$$

2.

$$\begin{array}{r} 34 \\ \times 11 \\ \hline \end{array}$$

3.

$$\begin{array}{r} 52 \\ \times 11 \\ \hline \end{array}$$

4.

$$\begin{array}{r} 96 \\ \times 11 \\ \hline \end{array}$$

5.

$$\begin{array}{r} 38 \\ \times 11 \\ \hline \end{array}$$

6.

$$\begin{array}{r} 69 \\ \times 11 \\ \hline \end{array}$$

7.

$$\begin{array}{r} 78 \\ \times 11 \\ \hline \end{array}$$

8.

$$\begin{array}{r} 83 \\ \times 11 \\ \hline \end{array}$$

9.

$$\begin{array}{r} 134 \\ \times 11 \\ \hline \end{array}$$

10.

$$\begin{array}{r} 205 \\ \times 11 \\ \hline \end{array}$$

11.

$$\begin{array}{r} 439 \\ \times 11 \\ \hline \end{array}$$

12.

$$\begin{array}{r} 516 \\ \times 11 \\ \hline \end{array}$$

13.

$$\begin{array}{r} 704 \\ \times 11 \\ \hline \end{array}$$

14.

$$\begin{array}{r} 659 \\ \times 11 \\ \hline \end{array}$$

15.

$$\begin{array}{r} 783 \\ \times 11 \\ \hline \end{array}$$

16.

$$\begin{array}{r} 801 \\ \times 11 \\ \hline \end{array}$$

17.

$$\begin{array}{r} 368 \\ \times 11 \\ \hline \end{array}$$

18.

$$\begin{array}{r} 736 \\ \times 11 \\ \hline \end{array}$$

19.

$$\begin{array}{r} 254 \\ \times 11 \\ \hline \end{array}$$

20.

$$\begin{array}{r} 327 \\ \times 11 \\ \hline \end{array}$$

Tip — Q1–8 use the 2-digit rule. Q9–20 use the 3-digit rule. Watch for carries when neighbour sums are 10 or more!



6.7 Multiply by 11 Part 2 · Q21–40

CHAPTER 6 PRACTICE



Bigger numbers, same trick!

Quick recap: copy the outer digits and put the **sum of every neighbouring pair** in between. Carry 1 to the left whenever a sum is ≥ 10 . Ready — set — solve!

21.

$$\begin{array}{r} 584 \\ \times 11 \\ \hline \\ \hline \end{array}$$

22.

$$\begin{array}{r} 937 \\ \times 11 \\ \hline \\ \hline \end{array}$$

23.

$$\begin{array}{r} 549 \\ \times 11 \\ \hline \\ \hline \end{array}$$

24.

$$\begin{array}{r} 627 \\ \times 11 \\ \hline \\ \hline \end{array}$$

25.

$$\begin{array}{r} 1725 \\ \times 11 \\ \hline \\ \hline \end{array}$$

26.

$$\begin{array}{r} 2843 \\ \times 11 \\ \hline \\ \hline \end{array}$$

27.

$$\begin{array}{r} 5219 \\ \times 11 \\ \hline \\ \hline \end{array}$$

28.

$$\begin{array}{r} 6054 \\ \times 11 \\ \hline \\ \hline \end{array}$$

29.

$$\begin{array}{r} 3247 \\ \times 11 \\ \hline \\ \hline \end{array}$$

30.

$$\begin{array}{r} 7823 \\ \times 11 \\ \hline \\ \hline \end{array}$$

31.

$$\begin{array}{r} 8509 \\ \times 11 \\ \hline \\ \hline \end{array}$$

32.

$$\begin{array}{r} 2564 \\ \times 11 \\ \hline \\ \hline \end{array}$$

33.

$$\begin{array}{r} 3012 \\ \times 11 \\ \hline \\ \hline \end{array}$$

34.

$$\begin{array}{r} 1546 \\ \times 11 \\ \hline \\ \hline \end{array}$$

35.

$$\begin{array}{r} 4157 \\ \times 11 \\ \hline \\ \hline \end{array}$$

36.

$$\begin{array}{r} 3258 \\ \times 11 \\ \hline \\ \hline \end{array}$$

37.

$$\begin{array}{r} 4120 \\ \times 11 \\ \hline \\ \hline \end{array}$$

38.

$$\begin{array}{r} 5274 \\ \times 11 \\ \hline \\ \hline \end{array}$$

39.

$$\begin{array}{r} 7218 \\ \times 11 \\ \hline \\ \hline \end{array}$$

40.

$$\begin{array}{r} 3608 \\ \times 11 \\ \hline \\ \hline \end{array}$$

Speed tip — With 4-digit numbers you get **three** neighbour-sums (between 4 digits). Work *right-to-left* so carries flow naturally into the next digit.



6.8 Multiply by 9, 99, 999 ... (Nikhilam)

CHAPTER 6

PRACTICE

9

Multiplication with equal digits of "9"

When the multiplier is made *entirely* of 9s (like 9, 99, 999, 9999), the answer appears almost by magic — using the sutra **Nikhilam Navataścaramaṁ Daśataḥ** ("all from 9, last from 10").

Describe the Method — $N \times$ (row of 9s of the same length)

Case 1 — same number of digits. For an n -digit number N multiplied by n nines:

1. **Left half** of the answer = $N - 1$
2. **Right half** of the answer = complement of N (i.e. $10^n - N$) — "all from 9, last from 10"
3. Join the two halves. Done!

$$88 \times 99$$

$$\text{Left} = 88 - 1 = 87$$

$$\text{Right} = 100 - 88 = 12$$

$$\text{Answer} = 8712$$

$$564 \times 999$$

$$\text{Left} = 564 - 1 = 563$$

$$\text{Right} = 1000 - 564 = 436$$

$$\text{Answer} = 563436$$

$$7 \times 9$$

$$\text{Left} = 7 - 1 = 6$$

$$\text{Right} = 10 - 7 = 3$$

$$\text{Answer} = 63$$

Handy shortcut for "all from 9, last from 10": subtract every digit from 9, except the **last** digit, which you subtract from 10. Example: complement of 564 = $(9-5)(9-6)(10-4) = 4\ 3\ 6 = 436$. ✓

1.

$$\begin{array}{r} 74 \\ \times 99 \\ \hline \end{array}$$

2.

$$\begin{array}{r} 82 \\ \times 99 \\ \hline \end{array}$$

3.

$$\begin{array}{r} 35 \\ \times 99 \\ \hline \end{array}$$

4.

$$\begin{array}{r} 68 \\ \times 99 \\ \hline \end{array}$$

5.

$$\begin{array}{r} 91 \\ \times 99 \\ \hline \end{array}$$

6.

$$\begin{array}{r} 62 \\ \times 99 \\ \hline \end{array}$$

7.

$$\begin{array}{r} 95 \\ \times 99 \\ \hline \end{array}$$

8.

$$\begin{array}{r} 38 \\ \times 99 \\ \hline \end{array}$$

9.

$$\begin{array}{r} 47 \\ \times 99 \\ \hline \end{array}$$

10.

$$\begin{array}{r} 18 \\ \times 99 \\ \hline \end{array}$$

11.

$$\begin{array}{r} 45 \\ \times 99 \\ \hline \end{array}$$

12.

$$\begin{array}{r} 89 \\ \times 99 \\ \hline \end{array}$$

13.

$$\begin{array}{r} 63 \\ \times 99 \\ \hline \end{array}$$

14.

$$\begin{array}{r} 77 \\ \times 99 \\ \hline \end{array}$$

15.

$$\begin{array}{r} 31 \\ \times 99 \\ \hline \end{array}$$

16.

$$\begin{array}{r} 58 \\ \times 99 \\ \hline \end{array}$$

17.

$$\begin{array}{r} 87 \\ \times 99 \\ \hline \end{array}$$

18.

$$\begin{array}{r} 54 \\ \times 99 \\ \hline \end{array}$$

19.

$$\begin{array}{r} 43 \\ \times 99 \\ \hline \end{array}$$

20.

$$\begin{array}{r} 96 \\ \times 99 \\ \hline \end{array}$$

21.

40

99

×

22.

37

99

×

23.

52

99

×

24.

70

99

×

Self-check tip — The answer of any 2-digit × 99 always has **4 digits**, and the **sum of all four digits is a multiple of 9**. Try it — $74 \times 99 = 7326 \rightarrow 7+3+2+6 = 18$. 🎯



6.8 Multiply by 999 & 9999 (Nikhilam · continued)

CHAPTER 6 PRACTICE



Same trick — bigger numbers!

The Nikhilam sutra scales up beautifully. 3-digit $\times$ 999 and 4-digit $\times$ 9999 use exactly the same "left = $N - 1$, right = complement" rule.

Quick recap — equal digits of 9

Left half = $N - 1$ · **Right half** = complement of N (all digits from 9, last digit from 10). Right half must have the **same number of digits** as N — pad with leading zeros if needed.

$$359 \times 999$$

$$\text{Left} = 358 \quad \cdot \quad \text{Right} = (9-3)(9-5)(10-9) = 641$$

$$\text{Answer} = 358641$$

$$5346 \times 9999$$

$$\text{Left} = 5345 \quad \cdot \quad \text{Right} = (9-5)(9-3)(9-4)(10-6) = 4654$$

$$\text{Answer} = 53454654$$

What if the multiplier has more 9s than N ?

Example 191×999 uses same rule (both are 3-digit vs 999). But 773×999 works too: $772 \mid 227 = 772227$. Just check digit counts match!

Special case — Q28 (674×999) and Q30 (999×989): In Q30 the multiplier 989 is *not* a row of 9s, so use classic column multiplication or Urdhva. Watch the "odd one out" — great puzzle!

25.

$$\begin{array}{r} 359 \\ \times 999 \\ \hline \hline \end{array}$$

26.

$$\begin{array}{r} 432 \\ \times 999 \\ \hline \hline \end{array}$$

27.

$$\begin{array}{r} 400 \\ \times 999 \\ \hline \hline \end{array}$$

28.

$$\begin{array}{r} 674 \\ \times 999 \\ \hline \hline \end{array}$$

29.

$$\begin{array}{r} 191 \\ \times 999 \\ \hline \hline \end{array}$$

30.

$$\begin{array}{r} 999 \\ \times 989 \\ \hline \hline \end{array}$$

31.

$$\begin{array}{r} 773 \\ \times 999 \\ \hline \hline \end{array}$$

32.

$$\begin{array}{r} 858 \\ \times 999 \\ \hline \hline \end{array}$$

33.

$$\begin{array}{r} 5346 \\ \times 9999 \\ \hline \hline \end{array}$$

34.

$$\begin{array}{r} 1827 \\ \times 9999 \\ \hline \hline \end{array}$$

35.

$$\begin{array}{r} 5083 \\ \times 9999 \\ \hline \hline \end{array}$$

36.

$$\begin{array}{r} 9999 \\ \times 9435 \\ \hline \hline \end{array}$$

37.

$$\begin{array}{r} 2850 \\ \times 9999 \\ \hline \hline \end{array}$$

38.

$$\begin{array}{r} 3948 \\ \times 9999 \\ \hline \hline \end{array}$$

39.

$$\begin{array}{r} 5714 \\ \times 9999 \\ \hline \hline \end{array}$$

40.

$$\begin{array}{r} 7070 \\ \times 9999 \\ \hline \hline \end{array}$$

Race your Scratch cat! — Build a tiny Scratch project that asks for a number and multiplies it by 999 using the sutra. Can you beat it with mental maths? Time yourself for extra fun. 🐱



6.9 Vedic Trick · Divide by 5, 25 & 50 Part 1 · Q1–20

CHAPTER 6 PRACTICE



Turn tricky division into easy multiplication!

Dividing by **5**, **25** or **50** looks scary — but each one is secretly a $\times$ in disguise. Learn one small rule and blaze through the practice.

Describe the Method — Divide by "half of 10, 100..."

The Vedic idea: any divisor that is a **friendly fraction of 10 or 100** can be flipped into a multiplication. Because **$5 = 10 \div 2$** , **$25 = 100 \div 4$** , and **$50 = 100 \div 2$** , we get three tiny rules:

- $\div 5 = \times 2$, then move the decimal **one** place left (*same as $\div 10$*)
- $\div 25 = \times 4$, then move the decimal **two** places left (*same as $\div 100$*)
- $\div 50 = \times 2$, then move the decimal **two** places left (*same as $\div 100$*)

$$35 \div 5$$

$$35 \times 2 = 70$$

$$70 \div 10 = 7$$

$$39 \div 5$$

$$39 \times 2 = 78$$

$$78 \div 10 = 7.8$$

$$75 \div 25$$

$$75 \times 4 = 300$$

$$300 \div 100 = 3$$

$$250 \div 50$$

$$250 \times 2 = 500$$

$$500 \div 100 = 5$$

Tip — If the answer has a decimal (like $39 \div 5 = 7.8$), you can also read it as a remainder: **7 remainder 4**. Both are correct — pick the form your teacher expects.

1.

$$\begin{array}{r} 35 \\ \div \quad 5 \\ \hline \end{array}$$

2.

$$\begin{array}{r} 20 \\ \div \quad 5 \\ \hline \end{array}$$

3.

$$\begin{array}{r} 45 \\ \div \quad 5 \\ \hline \end{array}$$

4.

$$\begin{array}{r} 80 \\ \div \quad 5 \\ \hline \end{array}$$

5.

$$\begin{array}{r} 95 \\ \div \quad 5 \\ \hline \end{array}$$

6.

$$\begin{array}{r} 70 \\ \div \quad 5 \\ \hline \end{array}$$

7.

$$\begin{array}{r} 125 \\ \div \quad 5 \\ \hline \end{array}$$

8.

$$\begin{array}{r} 190 \\ \div \quad 5 \\ \hline \end{array}$$

9.

$$\begin{array}{r} 39 \\ \div \quad 5 \\ \hline \end{array}$$

10.

$$\begin{array}{r} 76 \\ \div \quad 5 \\ \hline \end{array}$$

11.

$$\begin{array}{r} 83 \\ \div \quad 5 \\ \hline \end{array}$$

12.

$$\begin{array}{r} 56 \\ \div \quad 5 \\ \hline \end{array}$$

13.

$$\begin{array}{r} 91 \\ \div \quad 5 \\ \hline \end{array}$$

14.

$$\begin{array}{r} 49 \\ \div \quad 5 \\ \hline \end{array}$$

15.

$$\begin{array}{r} 87 \\ \div \quad 5 \\ \hline \end{array}$$

16.

$$\begin{array}{r} 66 \\ \div \quad 5 \\ \hline \end{array}$$

17.

$$\begin{array}{r} 75 \\ \div \quad 25 \\ \hline \end{array}$$

18.

$$\begin{array}{r} 150 \\ \div \quad 25 \\ \hline \end{array}$$

19.

$$\begin{array}{r} 225 \\ \div \quad 25 \\ \hline \end{array}$$

20.

$$\begin{array}{r} 375 \\ \div \quad 25 \\ \hline \end{array}$$

Tip — Q1–16 use the $\div 5$ rule ($\times 2$, then $\div 10$). Q17–20 switch to the $\div 25$ rule ($\times 4$, then $\div 100$). Watch out for decimals when the number isn't a clean multiple!



6.9 Divide by 25 & 50 Part 2 · Q21–40

CHAPTER 6

PRACTICE



Bigger numbers, same magic!

Quick recap — $\div 25 = \times 4$ then $\div 100$. $\div 50 = \times 2$ then $\div 100$. Move the decimal *two* places for both.

Ready — set — divide!

21.

$$\begin{array}{r} 850 \\ \div \quad 25 \\ \hline \hline \end{array}$$

22.

$$\begin{array}{r} 925 \\ \div \quad 25 \\ \hline \hline \end{array}$$

23.

$$\begin{array}{r} 1175 \\ \div \quad 25 \\ \hline \hline \end{array}$$

24.

$$\begin{array}{r} 1650 \\ \div \quad 25 \\ \hline \hline \end{array}$$

25.

$$\begin{array}{r} 217 \\ \div \quad 25 \\ \hline \hline \end{array}$$

26.

$$\begin{array}{r} 364 \\ \div \quad 25 \\ \hline \hline \end{array}$$

27.

$$\begin{array}{r} 510 \\ \div \quad 25 \\ \hline \hline \end{array}$$

28.

$$\begin{array}{r} 321 \\ \div \quad 25 \\ \hline \hline \end{array}$$

29.

$$\begin{array}{r} 187 \\ \div \quad 25 \\ \hline \hline \end{array}$$

30.

$$\begin{array}{r} 462 \\ \div \quad 25 \\ \hline \hline \end{array}$$

31.

$$\begin{array}{r} 531 \\ \div \quad 25 \\ \hline \hline \end{array}$$

32.

$$\begin{array}{r} 278 \\ \div \quad 25 \\ \hline \hline \end{array}$$

33.

$$\begin{array}{r} 250 \\ \div \quad 50 \\ \hline \hline \end{array}$$

34.

$$\begin{array}{r} 400 \\ \div \quad 50 \\ \hline \hline \end{array}$$

35.

$$\begin{array}{r} 650 \\ \div \quad 50 \\ \hline \hline \end{array}$$

36.

$$\begin{array}{r} 950 \\ \div \quad 50 \\ \hline \hline \end{array}$$

37.

$$\begin{array}{r} 1550 \\ \div \quad 50 \\ \hline \hline \end{array}$$

38.

$$\begin{array}{r} 1900 \\ \div \quad 50 \\ \hline \hline \end{array}$$

39.

$$\begin{array}{r} 2350 \\ \div \quad 50 \\ \hline \hline \end{array}$$

40.

$$\begin{array}{r} 5407 \\ \div \quad 50 \\ \hline \hline \end{array}$$

Self-check tip — Any number that ends in **0, 25, 50 or 75** divides *cleanly* by 25. Any number ending in **0** or **50** divides cleanly by 50. If your answer has a decimal, you did the maths right — the number simply isn't a perfect multiple!



6.10 Exercises · Operators & Maths

CHAPTER 6
PRACTICE



Crunch the numbers!

The green family is your maths toolbox. Time to wield it like a pro.

1 **CALCULATE** Solve mentally — then check in Scratch.

Given: **a = 2, b = 3, c = 4, d = 5**. Find the value of:

Expression	Working	Answer
(a + b)		
(c × d)		
(a × d) + (b × c)		
(a + b) × (c + d)		

2 **BLOCK EXPRESSION** Draw the green block stack.

How would you write **(5 × 2) + 3** using Scratch operator blocks? Sketch the block in the box (or describe which blocks go inside which).

Hint: nest a green × block *inside* a green + block.

3 **BUILD IT!** Vedic Calculator — Squares ending in 5

The sutra **Ekadhikena Purvena** says: **(N5)² = N × (N+1)** followed by **25**. Example: **35² = 3 × 4 = 12 ⇒ 1225**.

- Build a Scratch script that asks for a number **N** (like 3, 4, 5...).
- Compute **front = N × (N + 1)**.
- Use **join** to display **front** followed by **25**.

Write the answers your project should give for these inputs:

Input N	front (N×(N+1))	Answer (join with 25)
3 (so 35 ²)		
7 (so 75 ²)		
9 (so 95 ²)		

☐ My Vedic Squarer works!

Tip — Inside Scratch, big maths is just *small green blocks stacked inside each other*. Build slowly, test each step.



Worksheet · Operators & Math Blocks

CHAPTER 6 WORKSHEET



Operatos Challenge

Solve this worksheet to practise if/else decisions, Math logic.

- 1 A factory produced 15,678 toys in January and 23,945 toys in February. What is the total number of toys produced in both months?
 - a. 39,623
 - b. 38,513
 - c. 39,723
 - d. 38,613
- 2 The sum of two numbers is 72,500. If one number is 35,875, what is the other number?
 - a. 36,625
 - b. 37,375
 - c. 36,525
 - d. 37,625
- 3 A farmer harvested 12,450 kg of wheat. He sold 8,765 kg. How much wheat is left with him?
 - a. 3,685
 - b. 3,685 kg
 - c. 3,785 kg
 - d. 4,215 kg
- 4 Multiply 543 by 27
 - a. 14,661
 - b. 14,561
 - c. 14,671
 - d. 14,571
- 5 A box contains 144 pencils. If a school needs 35 such boxes, how many pencils will they have in total?
 - a. 5,040
 - b. 4,940
 - c. 5,140
 - d. 4,840
- 6 Divide 987 by 6. What is the quotient and the remainder?
 - a. Quotient = 164, Remainder = 3
 - b. Quotient = 160, Remainder = 7
 - c. Quotient = 165, Remainder = 3
 - d. Quotient = 164, Remainder = 2
- 7 Which of the following statements is true based on the properties of multiplication?
 - a. $12 \times 0 = 12$
 - b. $1 \times 45 = 45$
 - c. $(5 \times 3) \times 2 = 5 \times 3 + 2$
 - d. $10 \times (2 + 3) = 10 \times 2 + 3$
- 8 If you have 480 marbles and you want to share them equally among 15 friends, how many marbles will each friend get?
 - a. 32
 - b. 30
 - c. 35
 - d. 28
- 9 Which of the following is an example of the commutative property of addition?
 - a. $(10 + 5) + 3 = 10 + (5 + 3)$
 - b. $15 + 0 = 15$
 - c. $8 + 7 = 7 + 8$
 - d. $6 + 6 = 12$
- 10 A library has 2,500 books. On Monday, 150 books were borrowed, and on Tuesday, 120 books were returned. How many books are in the library now?
 - a. 2,470
 - b. 2,530
 - c. 2,230
 - d. 2,380

PRACTICE PLAN

How to use this worksheet

1. For each question, follow only the branch that matches the situation.

2. Check the answer key after solving, then rewrite two tricky rules as Scratch if/else blocks.



Worksheet • Operators & Math Blocks

CHAPTER 6

WORKSHEET

- 11** What is the product of the largest 3-digit number and the smallest 2-digit number?

a. 9,990 b. 99,900
c. 999 d. 99,900

12 If a bakery bakes 200 cookies every day, how many cookies will it bake in the month of April (30 days)?

a. 6,000 b. 6,200
c. 5,800 d. 6,100

13 A group of 18 students collected 720 stamps. If they shared them equally, how many stamps did each student get?

a. 40 b. 45
c. 30 d. 35

14 What is the value of $(54 \div 9) \times 7 - 12$?

a. 30 b. 42
c. 24 d. 36

15 If a car travels 65 km in one hour, how far will it travel in 18 hours?

a. 1,170 km b. 1,070 km
c. 1,270 km d. 1,150 km

16 Find the missing number: $7,500 \text{ } \{\rule{1cm}{0.4pt}\} 3,250$

a. 4,250 b. 4,350
c. 3,250 d. 4,150

17 17. A school bought 25 packets of chalk. Each packet contains 12 chalk sticks. If they already had 50 chalk sticks, how many chalk sticks do they have in total now?

a. 300 b. 350
c. 325 d. 250

18 What is the result when you divide the smallest 4-digit number by the smallest 1-digit number (excluding zero)?

a. 100 b. 1,000
c. 10 d. 1

19 A train carried 456 passengers. At the next station, 120 passengers got off, and 75 passengers got on. How many passengers are now on the train?

a. 411 b. 401
c. 381 d. 501

20 Which of the following operations has a product that ends in zero?

a. 23×15 b. 41×12
c. 17×13 d. 39×11

PRACTICE PLAN

How to use this worksheet

1. For each question, follow only the branch that matches the situation.
2. Check the answer key after solving, then rewrite two tricky rules as Scratch if/else blocks.

FOR SELF-CHECK, PARENTS & TEACHERS



Answer Key

Solutions to every exercise from Chapters 2 to 6. Try the questions first on your own — then peek here to check your work. The struggle is where the real learning happens!

CH 2 · SEQUENCING

CH 3 · LOOPS

CH 4 · VARIABLES

CH 5 · CONDITIONALS

CH 6 · OPERATORS



Answers · Sequencing & Events

CHAPTER 2 ANSWER KEY



Sequencing & Events — Solutions

Try the questions first. *Then* peek here to check.

1

PREDICT

Order on the stage

Blocks always run **top to bottom**. So the order is: **move 30** → **say "Hello!"** → **play meow** → **turn 15°**.

Order	What happens
1	Cat moves 30 steps
2	Cat says "Hello!"
3	Cat plays meow sound
4	Cat turns 15°

2

FIX THE BUG

Correct stack

Riya wants say "Hi" first, *then* jump. Correct order:

when clicked

say **Hi!** for **1** sec

move **50** steps

3

BUILD IT!

Birthday Sprite

The Events block that starts your script when the space-bar is pressed:

when [space] key pressed (yellow Events block, with the dropdown set to *space*)

4

LOGICAL REASONING

1.	B	2.	C	3.	A	4.	B	5.	A
6.	B	7.	B	8.	A	9.	B	10.	A
11.	B	12.	A	13.	B	14.	B	15.	C
16.	C	17.	B	18.	B	19.	A	20.	B

Remember — the block at the **top of the stack** always runs *first*. Change the order, change the story.

Pranav



Answers · Loops

CHAPTER 3
ANSWER KEY



Loops — Solutions

One loop block can do the work of dozens. Check your counts below.

1 COUNT IT How many times?

Loop	Inside runs how many times?
repeat 10	10 times
repeat 25	25 times
forever	Forever — until you press the red ■ (stop)
repeat 4 inside repeat 3	12 times (outer 3 × inner 4)

2 MATCH THE SHAPE repeat × turn = ?

Rule: turn = $360^\circ \div \text{number of sides}$.

Shape	repeat	turn (degrees)
Triangle	3	120°
Square	4	90°
Pentagon	5	72°
Octagon	8	45°

3 BUILD IT! Rocket Countdown

To make the counter go down from 10 to 1, use:

change counter by -1 (negative one — orange Variables block)

4 LOGICAL REASONING

1.	B	2.	A	3.	B	4.	B	5.	A
6.	C	7.	B	8.	B	9.	B	10.	D
11.	B	12.	A	13.	A	14.	B	15.	B
16.	B	17.	A	18.	B	19.	D	20.	A

Remember — a loop is just "do this again". If you're copying blocks more than twice — switch to a loop.



Answers · Variables & Lists

CHAPTER 4 ANSWER KEY



Variables & Lists — Solutions

Trace each step in order. Remember: **set** replaces, **change** adds to.

1

TRACE IT

What's in the score box?

Start: **score** = 0.

Instruction	score is now...
change score by 5	5
change score by 3	8
set score to 10	10 (set replaces!)
change score by -4	6
change score by 1	7

2

LIST TRACE

Final list

Trace: [apple] → [apple, banana] → [apple, banana, cherry] → **delete item 1** → [banana, cherry] → [banana, cherry, date].

Final **fruits** list (only 3 items remain):

1	banana	2	cherry	3	date	4	— empty —
---	--------	---	--------	---	------	---	-----------

3

BUILD IT!

Friend Collector

Outline of a correct project:

1. Make a list called **friends**.
2. Use **repeat 3** with: *ask "Friend's name?" and wait*, then *add (answer) to friends*.
3. Use another **repeat 3** with a counter **i**: *say (item i of friends) for 1 sec*, then *change i by 1*.

4

LOGICAL REASONING

1.	A	2.	B	3.	B	4.	C	5.	A
6.	C	7.	B	8.	B	9.	C	10.	C
11.	B	12.	B	13.	C	14.	A	15.	A
16.	B	17.	B	18.	A	19.	B	20.	A

Remember — deleting item 1 always shifts every other item *up* by one. That's why the final list is only 3 long, not 4.



Answers · Conditionals

CHAPTER 5 ANSWER KEY



Conditionals — Solutions

Every condition is a question with a **yes/no** answer. Let's check yours.

1

TRUE OR FALSE?

Answers

Condition	Answer
$10 > 5$	True
$7 = 7$	True
"cat" = "Cat"	True in Scratch (Scratch ignores capital letters in =). In most other coding languages this would be False .
$100 < 99$	False
$(3 + 2) = 5$	True

2

PREDICT

What does the cat say?

Condition: **answer > 10**. If true → "Big!", else → "Small!".

If you type...	Cat says...
15	Big! (15 > 10 is true)
10	Small! (10 > 10 is false — > means <i>strictly</i> greater)
3	Small!
100	Big!

3

BUILD IT!

Pass / Fail Reporter

You need **2 nested if-else blocks**:

- Outer: **if** answer ≥ 90 → say "Excellent! 🌟" **else** ...
- Inner (in the else): **if** answer ≥ 50 → say "Pass 😊" **else** say "Try again 🙄".

4

LOGICAL REASONING

1.	A	2.	B	3.	C	4.	C	5.	C
6.	A	7.	B	8.	C	9.	B	10.	B
11.	C	12.	C	13.	B	14.	A	15.	A
16.	B	17.	A	18.	B	19.	B	20.	A

Remember — the symbol $>$ is *strictly greater than*. For "10 or more", use $\geq$ (or check **not (answer < 10)**).



Operators & Maths — Solutions

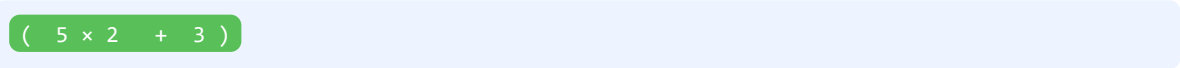
BODMAS still rules — even when you're stacking green blocks.

1 **CALCULATE** a = 2, b = 3, c = 4, d = 5

Expression	Working	Answer
(a + b)	2 + 3	5
(c × d)	4 × 5	20
(a × d) + (b × c)	(2×5) + (3×4) = 10 + 12	22
(a + b) × (c + d)	(2+3) × (4+5) = 5 × 9	45

2 **BLOCK EXPRESSION** (5 × 2) + 3

Nest a green × **block** *inside* a green + **block**:



The final answer is (5 × 2) + 3 = 10 + 3 = 13.

3 **BUILD IT!** Vedic Calculator — Squares ending in 5

Rule: **front** = N × (N + 1), then **join** with 25.

Input N	front (N × (N+1))	Answer (join with 25)
3 (35 ²)	3 × 4 = 12	1225
7 (75 ²)	7 × 8 = 56	5625
9 (95 ²)	9 × 10 = 90	9025

4 **LOGICAL REASONING**

1.	C	2.	A	3.	A	4.	A	5.	A
6.	A	7.	B	8.	A	9.	C	10.	A
11.	A	12.	A	13.	A	14.	A	15.	A
16.	A	17.	B	18.	B	19.	A	20.	A

Remember — *Ekadhikena Purvena* means "**by one more than the previous one**". For 35², the previous digit is 3, so we multiply 3 by **one more** = 4. Then stick 25 on the end. Magic!

BONUS CAPSTONE · TAKE-HOME PROJECT



The *Water Cycle* Animation

*You have learned events, loops, variables, conditionals, operators and Vedic sutras. Now mix them all together to build a real **Science animation** — the journey of a single water drop from the sea to the sky and back as rain.*

5 SPRITES

BROADCASTS

CLONING NEW

VARIABLES

REAL SCIENCE

SHOW AT HOME 🏠



Capstone · The Water Cycle

HOMEWORK

BONUS

TAKE-HOME PROJECT

What we're building



An animated Science scene that shows the **three big stages** of the water cycle, looping forever:

1. Evaporation — the Sun heats the sea, a tiny water drop rises into the sky. **2. Condensation** — many drops meet and the cloud grows bigger. **3. Precipitation** — when the cloud is heavy enough, it rains.

Meet your 5 sprites + 1 boss

Stage (the boss)

Doesn't move — but **tells everyone what to do**. Owns the master loop and two variables: **Cycle** and **drops**.

Sun

Just pulses gently in the top-right corner. Decorative — no broadcasts from here.

WaterDrop

Hides at the bottom. When it hears "evaporate", it shows up, says "Evaporation!" and glides up to the cloud.

Cloud

Sits at the top. Grows bigger every "condense". Resets to its small size after every rainstorm.

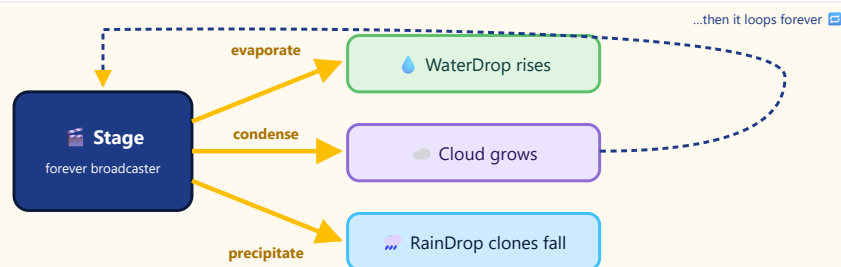
RainDrop

Hides on flag. When it hears "precipitate", it **clones itself 8 times** and the clones fall to the ground.

Variables

Cycle — how many full water cycles we've finished.
drops — how many drops have collected in the cloud this round.

How the sprites talk — the broadcast flow



The Stage is the only sprite that broadcasts. WaterDrop, Cloud and RainDrop just **listen** and react.



Step 1–3 · Backdrop, Sun & WaterDrop

BONUS
TAKE-HOME PROJECT

Step 1 · Paint the backdrop 🎨

Click the **Stage** (right side) → **Backdrops** tab → **Paint**. Make a sky that fades from blue at the top to sand at the bottom, with a dark blue sea at the bottom. This is the world your animation lives in.

Step 2 · The Sun — gentle pulse 🌞

Add a Sun sprite (paint a yellow circle). Place it at **x = 180, y = 130**. The Sun just **looks** alive — it doesn't control the cycle.

SUN SPRITE

```

when clicked
  go to x: 180 y: 130
  set size to 80 %
  forever
    change size by 12
    wait 0.25 secs
    change size by -12
    wait 0.25 secs
  
```

💡 Why not let the Sun broadcast?

If the Sun shouted **evaporate!** every half-second, it would *restart* the WaterDrop's script before the drop had time to glide. The Sun must stay **decorative**. The full secret is on the next page.

Step 3 · The WaterDrop — rises on cue 💧

Paint a small blue tear-drop sprite. It needs **two scripts**: hide at the start, and rise when it hears "evaporate".

Script A · Start hidden

WATERDROP SPRITE

```

when clicked
  hide
  
```

Script B · When I hear "evaporate"

WATERDROP SPRITE · RECEIVES BROADCAST

```

when I receive evaporate
  go to x: pick random -200 to 200 y: -120
  show
  say Evaporation! for 0.5 secs
  glide 1.5 secs to x: -100 y: 100
  hide
  
```


Evaporation

Heat from the Sun gives water molecules so much energy that they **turn into vapour** and rise into the sky. Our drop's upward glide shows exactly that.

**Glide = animation**

The **glide** block moves the sprite smoothly from one (x, y) to another in a chosen time. Perfect for showing motion in nature.



Step 4–5 · Cloud & RainDrop

BONUS
TAKE-HOME PROJECT

Step 4 · The Cloud — three short scripts

Paint a fluffy white cloud. Place it at **x = -100, y = 100**. The Cloud listens to **two** broadcasts and resets itself with one start-up script.

A · Set up

CLOUD · ON FLAG

```
when clicked
  go to x: -100 y: 100
  set size to 60 %
  show
```

B · On "condense"

CLOUD GROWS

```
when I receive condense
  say Condensation! for 0.5 secs
  change size by 15
```

C · On "precipitate"

CLOUD RESETS

```
when I receive precipitate
  say Rain! for 0.5 secs
  set size to 60 %
```

Step 5 · The RainDrop — meet cloning NEW

A **clone** is a copy of a sprite that Scratch creates while the project is running. Perfect for rain: one tiny RainDrop sprite can become *eight* falling drops.

A · Hide at start

RAINDROP · MAIN

```
when clicked
  hide
```

B · Spawn 8 clones

ON "PRECIPITATE"

```
when I receive precipitate
  repeat 8
    create clone of myself
    wait 0.08 secs
```

C · Each clone falls

CLONE BEHAVIOUR

```
when I start as a clone
  go to x: pick random -140 to -60 y: 120
  show
  glide 0.8 secs to x: x position y: -120
  delete this clone
```

Why delete this clone?

Without it, every cycle would leave 8 more drops frozen at the bottom of the stage. After hundreds of cycles your Scratch project would slow down. **Delete this clone** = good housekeeping. Always clean up your clones!



Step 6 · The Stage Drives the Cycle

BONUS

TAKE-HOME PROJECT

Click on the **Stage** (right side, below the green flag). The Stage gets its *own* script area. Make **two variables** for *this sprite only*: **Cycle** (starts at 0) and **drops** (starts at 0). These belong to the whole project.

The master loop

STAGE · THE CONDUCTOR OF THE ORCHESTRA



⚠ The "broadcast and wait" trap

The bug we have to avoid

If you use the plain **broadcast** block in a fast **forever** loop, Scratch will fire *evaporate* again **before** the WaterDrop finishes gliding — and Scratch **restarts** the receiving script from the top. Result: you only ever see "Evaporation!" — the drop never reaches the cloud. 🤖

The fix — use **broadcast ... and wait**. The Stage now *pauses* until the WaterDrop has finished its glide, then fires the next message. Each phase gets to finish. Beautiful!

Timeline — what happens in one cycle 🕒

Time	Who acts	What you see
0.0 s	Stage → "evaporate"	WaterDrop appears at sea, says "Evaporation!"
0.5–2.0 s	WaterDrop	Glides up to the cloud, hides
2.0 s	Stage	drops changes from 0 → 1 → 2 → 3 (one per cycle)
2.0 s	Stage → "condense"	Cloud says "Condensation!" and grows by 15%
~3.0 s	Stage checks drops > 2	After the 3rd drop, the condition is true → time to rain!
~3.5 s	Stage → "precipitate"	8 RainDrop clones fall, Cloud resets to small, Cycle goes up by 1
loop	Stage	Back to "evaporate" — the water cycle continues 🔄



Run It · Tweak & Extend

BONUS
TAKE-HOME PROJECT

What you should see on the stage 🎬



Cycle = 3 means the water has already gone sea → sky → rain → sea **three times**. drops = 2 means one more evaporation will trigger the next rainfall.

★ Bonus challenges — try at least one!

🌈 Add a rainbow

After "precipitate", broadcast **rainbow** — and let a new Rainbow sprite **show** for 2 seconds, then **hide**.

❄️ Make it snow

Add a Temperature variable. When Temperature < 0, broadcast **snow** instead of precipitate. Use a white snowflake costume.

🔊 Add sound effects

From the **Sounds** tab, add a "drip" sound to RainDrop and play it when each clone starts.

🏔️ Add a mountain

Paint a brown mountain on the backdrop. Where do **rivers** fit into the water cycle?

YOUR TURN · ACTIVITY B.1

Plan, then build

1. Open Scratch (online: scratch.mit.edu · offline: Scratch Desktop). Start a new project named **"Water Cycle"**.
2. Make all 5 sprites + 2 Stage variables (**Cycle** and **drops**).
3. Type the scripts from pages B2 → B4. Test by clicking the green flag.
4. Bonus 🌟: Pick **one** challenge from above and add it. Tick it here once it works:

- ☐ 🌈 Rainbow
 ☐ ❄️ Snow
 ☐ 🔊 Sound
 ☐ 🏔️ Mountain
- ☐ I showed my finished project to a parent or friend!

**ASK A.I. Stuck? Try this prompt —**

In Scratch, my **Stage** uses a **forever** loop with **broadcast evaporate and wait** → **broadcast condense and wait** → if **drops > 2** then **broadcast precipitate and wait**. My **WaterDrop** hears "evaporate", says "*Evaporation!*" and glides up — but the cloud never grows. What might I have done wrong, and what should I check first?

💡 Telling A.I. the **order of broadcasts** and which *sprite listens to which one helps it find the bug fast.*

WELL DONE, CODER!



You finished *Level 1 Young Explorer*

*From your first sprite to your own Vedic calculator — every block you snapped, every sutra you used, every bug you fixed has made you a stronger thinker. See you in **Volume 2 · Builders** for Grade 5!*

SEQUENCING ✓

LOOPS ✓

VARIABLES & LISTS ✓

CONDITIONALS ✓

OPERATORS ✓

VEDIC SUTRAS ✓

SELF-PRACTICE ✓

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– Albert Einstein

”



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